



SMART GYM MANAGEMENT SYSTEM

A Full-Stack Web Application for Comprehensive Gym Operations Management
(React + Spring Boot + Oracle)

Submitted by:

SUTHAR ARYAN SUJALKUMAR

Enrollment No: 23C25512

Under the guidance of

Dr. Ashutosh Abhangi

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School of Computer Science Engineering & Technology

ITM SLS Baroda University

Paldi Village, Vadodara – 390510



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Enrolment Number: 23C25512
Programme: B.Tech (Information Technology)
Semester: Semester 8
Internship Period: 17th November 2025 – 17th May 2026
Organisation: Bharti Soft Tech Pvt. Ltd.

Student Signature: _____ **Date:** _____

Supervisor: Dr. Ashutosh Abhangi

Supervisor Signature: _____ **Date:** _____

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Enrolment Number: 23C25512

Programme: B.Tech (Information Technology)

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Abstract

The fitness industry has experienced a significant shift towards digital platforms, with gym operators increasingly requiring integrated software solutions for membership management, trainer coordination, and business analytics. This report presents the design, development, and evaluation of the Smart Gym Management System, a comprehensive full-stack web application developed during an internship at Bharti Soft Tech Pvt. Ltd. under the supervision of Dr. Ashutosh Abhangi.

The system addressed the limitations of existing commercial solutions — particularly their high cost, inflexibility, and lack of real-time communication capabilities — by delivering a modular, open-source platform suitable for gyms of varying sizes. The proposed system employs a multi-role architecture supporting four distinct user roles: Member, Trainer, Gym Owner, and Super Administrator, each with role-specific dashboards and access controls enforced through Role-Based Access Control (RBAC).

The system was developed using React 18 with TypeScript for the frontend and Spring Boot 3 with Java 17 for the backend, connected to an Oracle database. Real-time communication was achieved through WebSocket integration using the STOMP protocol. Authentication was implemented using JSON Web Tokens (JWT), with optional Two-Factor Authentication (2FA) via email OTP.

Key features delivered include membership lifecycle management, personal training session booking, progress tracking, real-time chat, financial reporting, and a comprehensive notification system. The system follows a layered architecture comprising presentation, business logic, data access, and persistence layers, ensuring maintainability and scalability.

Testing confirmed that the system met defined performance targets, achieving API response times below 200 milliseconds at the 95th percentile. Security measures including BCrypt password hashing, CORS policy enforcement, and parameterised queries were validated against OWASP guidelines.

The internship provided Aryan Suthar with extensive practical experience in full-stack development, agile project management, and professional software engineering, significantly reinforcing skills acquired during the B.Tech (IT) programme.

Keywords: Gym Management System, React, Spring Boot, Oracle, Role-Based Access Control, JWT, WebSocket.

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Chapter 3	Weekly Workflow Schedule
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Chapter 5	Entity-Relationship (ER) Diagram
Chapter 6	UML Class Diagram; UML Sequence Diagrams
Chapter 7	Data Flow Diagrams — Level 0 (Context) and Level 1
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List of Symbols

Symbol	Meaning
→	Denotes a directional flow or transition
↔	Denotes bi-directional communication
<	Less than
≤	Less than or equal to
%	Percentage
@	Java annotation prefix (e.g. @Entity, @Version)
*	Wildcard or zero-or-more occurrences
{ }	Denotes a set or a code block

List of Abbreviations

Abbreviation	Definition
2FA	Two-Factor Authentication
ACID	Atomicity, Consistency, Isolation, Durability
API	Application Programming Interface
BCrypt	Blowfish Crypt (password hashing algorithm)
CORS	Cross-Origin Resource Sharing
CRM	Customer Relationship Management
CSS	Cascading Style Sheets
CRUD	Create, Read, Update, Delete
DB	Database
DTO	Data Transfer Object
E2E	End-to-End (Testing)
ER	Entity-Relationship
GDPR	General Data Protection Regulation
HTML	HyperText Markup Language
HTTP	HyperText Transfer Protocol
HTTPS	HyperText Transfer Protocol Secure
IDE	Integrated Development Environment
JPA	Java Persistence API
JSON	JavaScript Object Notation
JWT	JSON Web Token
MVC	Model-View-Controller
NIST	National Institute of Standards and Technology
OAuth	Open Authorisation
ORM	Object-Relational Mapping
OTP	One-Time Password
OWASP	Open Web Application Security Project
PCI-DSS	Payment Card Industry Data Security Standard
POS	Point of Sale

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Abbreviation	Definition
PT	Personal Training
RBAC	Role-Based Access Control
REST	Representational State Transfer
RFC	Request for Comments
SaaS	Software as a Service
SMS	Short Message Service
SQL	Structured Query Language
STOMP	Simple Text Oriented Messaging Protocol
TLS	Transport Layer Security
TOC	Table of Contents
UI	User Interface
UML	Unified Modelling Language
URL	Uniform Resource Locator
UX	User Experience
WCAG	Web Content Accessibility Guidelines
WS	WebSocket
XSS	Cross-Site Scripting

Chapter 1: Introduction

1.1 Background of the Organisation

Bharti Soft Tech Pvt. Ltd. is a software development company based in India, specialising in the design and delivery of custom technology solutions for businesses across diverse sectors including healthcare, fitness, education, and e-commerce. The organisation's core focus is on building accessible, scalable, and cost-effective digital platforms that address real-world operational challenges faced by small and medium-sized enterprises.

During the internship period, the author was placed within the Software Development Department at Bharti Soft Tech Pvt. Ltd. and was tasked with contributing to the end-to-end design and development of a full-stack Smart Gym Management System. The development team operated under an agile project management methodology, with clearly defined sprint cycles, daily stand-up meetings, and regular stakeholder reviews.

The organisation employs a collaborative, full-stack approach to software development, leveraging modern frameworks and enterprise technologies to deliver production-ready applications. Under the supervision of Dr. Ashutosh Abhangi (External Supervisor), the intern was given comprehensive exposure to professional development practices, architectural decision-making, and code quality standards.

1.2 Objectives of the Organisation

The primary objectives of Bharti Soft Tech Pvt. Ltd. are as follows:

1. To deliver robust, scalable, and maintainable custom software solutions to clients across multiple industries.
2. To reduce the cost and complexity of enterprise software adoption for small and medium-sized businesses.
3. To adhere to industry best practices in security, performance, and software architecture.
4. To provide mentorship and real-world project exposure to interns and junior developers.
5. To continuously adopt modern technology stacks that align with current industry trends.

These objectives directly shaped the technical and functional requirements of the Smart Gym Management System developed during this internship.

1.3 Products and Main Services of the Organisation

Bharti Soft Tech Pvt. Ltd. offers the following core products and services:

Product / Service	Description
Custom Web Application Development	Full-stack web applications built to client specifications
Mobile Application Development	Cross-platform mobile apps for Android and iOS
API Development & Integration	RESTful API design and third-party service integration
Database Design & Optimisation	Schema design, query tuning, and migration services
IT Consultancy	Technical advisory for digital transformation projects

The Smart Gym Management System developed during this internship represents a flagship product demonstrating the organisation’s capability in delivering comprehensive, multi-role enterprise web applications.

1.4 Organisation Structure and Workflow

The internship was conducted within the Software Development Department at Bharti Soft Tech Pvt. Ltd. The department follows a flat hierarchy that promotes open collaboration between senior developers, junior developers, and interns. The author worked under the direct supervision of Dr. Ashutosh Abhangi throughout the placement.

The development workflow followed the Agile Scrum methodology:

- **Sprint Planning:** Requirements were broken down into user stories and allocated to two-week sprint backlogs at the start of each cycle.
- **Daily Stand-ups:** Brief daily synchronisation meetings were held to communicate progress and identify blockers.
- **Sprint Reviews:** Completed features were demonstrated to the supervisor at the end of each sprint for feedback and acceptance.
- **Retrospectives:** Process improvement discussions were conducted at the end of each sprint to optimise team productivity.

Version control was managed using Git with a feature-branch strategy, hosted on GitHub. Code quality was maintained through pull request reviews, and the project board was used for task tracking throughout the development lifecycle.

Chapter 2: Literature Review — Project Development Journey

2.1 Introduction

The fitness industry has undergone profound digital transformation over the past decade. Contemporary gym management platforms have evolved from basic membership-tracking tools into comprehensive, multi-role software ecosystems that integrate scheduling, real-time communication, financial reporting, and business analytics [12]. Despite the abundance of commercial solutions such as Mindbody, Zen Planner, PushPress, and GymMaster, a critical gap persists: most proprietary platforms impose high subscription costs — typically between USD 85 and USD 159 per month — are closed-source, and offer limited customisation [14]. Independent and small-to-medium gym operators are disproportionately affected by these constraints.

This chapter documents the complete development journey of the Smart Gym Management System (hereafter referred to as *AthlonX* or the *system*) undertaken during the internship at Bharti Soft Tech Pvt. Ltd. under the supervision of Dr. Ashutosh Abhangi. The development was structured across four formal review phases, each corresponding to a distinct set of objectives, deliverables, and learning milestones. The narrative presented here draws on the project planning documents, product requirements documents (PRDs), design decisions, and working code artefacts produced throughout the internship period.

2.2 Review Phase 1: Project Inception, Requirements Analysis, and Foundation Building

2.2.1 Project Initiation and Stakeholder Alignment

The project commenced with an intensive requirements-gathering and feasibility analysis phase (Weeks 1–4 of the internship). Under the guidance of Dr. Ashutosh Abhangi, the author conducted a structured literature search examining existing commercial gym management solutions to identify functional gaps and motivate the development of a purpose-built system. A comparative study of leading platforms — Mindbody, Zen Planner, PushPress, GymMaster, and Wodify — was conducted and documented in the project’s analysis plan.

The key deficiencies identified across commercial platforms are summarised below:

System	Key Strengths	Critical Limitations
Mindbody	Booking, payments, marketing	High cost (USD 139+/mo), complex setup
Zen Planner	Membership, billing, re- porting	Limited customisation
GymMaster	Access control, POS, member app	Desktop-focused, poor web UX
PushPress	CRM, automation, mobile app	Pricing tier restrictions (USD 79–159/mo)
Wodify	CrossFit focus, performance tracking	Niche market, limited general use

This gap analysis directly shaped the foundational vision of the AthlonX system: an open-source, self-hosted, cost-free alternative capable of matching or exceeding the feature depth of commercial competitors [14].

2.2.2 Technology Stack Selection and Rationale

A critical early decision concerned the selection of the technology stack. Following a structured evaluation, the following technologies were selected:

Frontend — React 18 with TypeScript: React’s component-based architecture enables the development of reusable, maintainable UI components, and its Virtual DOM optimises rendering performance for data-intensive dashboards [1]. TypeScript was adopted to add static type safety, which materially reduces runtime errors and improves the long-term maintainability of a large codebase [3]. Alternative frameworks, including Vue.js and Angular, were evaluated; React was selected for its maturity, enterprise adoption, and the richness of its ecosystem (React Router, Axios, Framer Motion).

Backend — Spring Boot 3 with Java 17: Spring Boot’s auto-configuration and opinionated defaults enabled rapid set-up of enterprise-grade REST APIs. Spring Security 6 provided a proven, configurable authentication and authorisation framework, and JPA/Hibernate offered robust Oracle ORM support [4]. Node.js/Express and Django were considered but rejected due to concurrency limitations and ecosystem maturity considerations for enterprise Oracle integration.

Database — Oracle Database: Oracle was mandated by the organisation owing to its ACID compliance, enterprise-grade auditing capabilities, and proven integration with Hibernate for complex entity-relationship mapping [18].

2.2.3 Initial Architecture Design

The system was designed around a layered, four-tier architecture comprising a Presentation Layer (React SPA), a Business Logic Layer (Spring Boot services), a Data Access Layer (Spring Data JPA repositories), and a Persistence Layer (Oracle Database). A multi-role access model was defined from the outset, identifying four distinct user roles: **Member**, **Trainer**, **Gym Owner**, and **Super Administrator**. Role-Based Access Control (RBAC) was designed to follow the NIST Core RBAC model [7], with hierarchical role inheritance enforced at both the API layer (via Spring Security's `@PreAuthorize` annotations) and the React frontend (via protected route components).

Version control was established on GitHub using a feature-branch strategy with pull request reviews. The Agile Scrum methodology was adopted for sprint planning, with two-week sprint cycles aligned with supervisor review meetings.

2.3 Review Phase 2: Core Feature Implementation and Backend Development

2.3.1 Authentication and Authorisation Subsystem

Phase 2 (Weeks 5–10) focused on implementing the core backend subsystems. The authentication layer was built using JSON Web Tokens (JWT) as specified in RFC 7519 [8], implemented through the JJWT library. The authentication flow comprised:

1. User credentials submitted via the login endpoint (`POST /api/auth/login`).
2. Server validates credentials against BCrypt-hashed passwords stored in the database, using 12 rounds as recommended by OWASP [11].
3. A short-lived Access Token (15 minutes) and a long-lived Refresh Token (7 days) are issued upon successful authentication.
4. Optional Two-Factor Authentication (2FA) via email OTP is triggered for accounts with 2FA enabled.
5. Subsequent requests carry the JWT in the `Authorization: Bearer` header; the filter chain validates the token and populates the Spring Security context.

This stateless authentication model enables horizontal scalability without server-side session storage, aligning with cloud-native deployment best practices [4].

2.3.2 Membership Management Module

The membership management module was identified as the core business feature of the system. The implementation delivered a complete membership lifecycle covering the following states: **Pending Ap-**

proval → **Active** → **Expired / Suspended**. Gym owners were provided with administrative endpoints to approve, reject, or suspend memberships. The module included:

- Package creation and pricing management by Gym Owners.
- Member-initiated subscription requests with automatic status transitions upon owner approval.
- Scheduled background jobs (via Spring's `@Scheduled` annotation) for automatic membership expiry detection and notification triggering.
- PT (Personal Training) session credit tracking, deducted upon trainer session completion.

A significant technical challenge arose from the need to handle concurrent membership status updates across multiple sessions. This was resolved through optimistic locking at the JPA entity level (`@Version` annotation), preventing lost updates in concurrent environments [33].

2.3.3 Personal Training Session Booking System

The personal training session booking system enabled members to browse trainer profiles, view available time slots, and submit booking requests. The backend enforced business rules including:

- A member must hold an active membership to book a PT session.
- Session credit balances are validated at the time of booking.
- Trainers receive real-time notifications upon new booking requests via the WebSocket notification subsystem.
- Session completion triggers credit deduction and allows the trainer to submit a session summary.

Database query optimisation was a focus during this phase. N+1 query problems, a common issue documented in ORM literature [33], were identified through query logging and resolved using JPA JOIN FETCH strategies and entity graph annotations. This optimisation contributed to achieving API response times below 200 milliseconds at the 95th percentile.

2.4 Review Phase 3: Frontend Development, Real-Time Features, and System Integration

2.4.1 Multi-Role Dashboard Implementation

Phase 3 (Weeks 11–16) was dedicated to the frontend implementation and the integration of real-time communication features. The React 18 SPA was structured around a role-aware routing system that delivered distinct dashboards for each of the four user roles. The design followed a component-based architecture, with all reusable UI elements — tables, modal dialogs, stat cards, progress charts — encapsulated as typed React components.

The design system was guided by modern web UI/UX principles as documented in human–computer interaction research [12]:

- **Owner Dashboard:** Real-time KPI cards (active members, monthly revenue, pending requests), financial analytics via Recharts 2.x bar and line charts, and trainer schedule views.
- **Trainer Dashboard:** Assigned client roster, upcoming session calendar, and client progress tracking with visual indicators.
- **Member Dashboard:** Membership status card, PT session credits, progress entry forms, and booking interface.
- **Super Admin Dashboard:** System-wide analytics, gym management, user moderation, error log viewer, and audit log access.

Axios 1.x was used for all HTTP communication, with request interceptors handling JWT injection and response interceptors managing token refresh flows and global error handling [26].

2.4.2 Real-Time WebSocket Integration

The real-time chat and notification subsystems were implemented using Spring WebSocket with the STOMP protocol for message routing over WebSocket connections. The WebSocket protocol (RFC 6455) enables bidirectional, low-latency communication, achieving observed latencies below 50 milliseconds, compared to polling-based approaches which typically incur delays exceeding 1,000 milliseconds [9].

The chat system allowed Members and Trainers to communicate in dedicated conversation threads. Key features delivered included:

- Message delivery via STOMP topic subscriptions.
- Real-time read-receipt updates.
- Presence indicators and unread message badge counters.
- Lazy-loaded conversation history with pagination.

A significant challenge encountered during this phase was the management of WebSocket connection state across page navigation in the React SPA. This was resolved using a dedicated `WebSocketContext` React context provider that maintained a persistent STOMP client instance throughout the application lifecycle.

2.4.3 Progress Tracking and Notification System

A member progress tracking module was implemented allowing members to log body metrics (weight, body fat, measurements) at regular intervals. Progress data was visualised using Recharts 2.x line and area charts [30]. The charting implementation drew on best practices for data visualisation in fitness applications as documented by Sharma et al. [12].

The notification system adopted an event-driven, observer-pattern architecture [10]. Domain events (membership approval, session booking, expiry warnings) triggered server-side notification creation, which were then delivered to connected clients via the WebSocket broadcast channel or stored for retrieval on the next login.

2.5 Review Phase 4: Security Hardening, Testing, and System Polish

2.5.1 Security Implementation and OWASP Compliance

Phase 4 (Weeks 17–20) concentrated on security hardening, testing, and system polish. The security architecture was validated against the OWASP Top Ten 2021 [21] vulnerabilities:

- **Injection (A03):** All database queries used parameterised JPA queries, eliminating SQL injection vectors.
- **Broken Authentication (A07):** JWT with HMAC-SHA512 signature, BCrypt password hashing (12 rounds), and 2FA collectively addressed authentication weaknesses.
- **Broken Access Control (A01):** Method-level `@PreAuthorize` annotations enforced RBAC at the service layer, preventing privilege escalation.
- **Security Misconfiguration (A05):** CORS policy was configured with an explicit whitelist of permitted origins, blocking cross-origin requests from unauthorised domains.
- **Cryptographic Failures (A02):** All sensitive data in transit was protected via HTTPS/TLS. Passwords were stored exclusively as BCrypt hashes, never in plaintext.

Password security followed the NIST Digital Identity Guidelines [22] and OWASP Password Storage recommendations [11], prioritising adaptive hashing algorithms resistant to brute-force attacks.

2.5.2 Testing and Quality Assurance

Unit testing was conducted using JUnit 5 and Mockito, following the AAA (Arrange, Act, Assert) pattern [10]. Test coverage focused on critical service-layer methods in the authentication, membership, and PT session modules. API endpoint correctness was validated using Postman collections and Newman for automated API regression testing.

Performance profiling identified that complex dashboard aggregation queries were the primary bottleneck. Materialised query results for frequently polled dashboard endpoints and strategic database indexing on high-cardinality foreign key columns (`user_id`, `gym_id`, `membership_id`) reduced query execution times from an average of 450 ms to below 80 ms [18].

2.5.3 Comparative Feature Analysis and Competitive Positioning

Upon the completion of Phase 4, a comprehensive feature comparison was conducted against the benchmark commercial platforms:

Feature	AthlonX (This System)	Mindbody	PushPress	Zen Planner
Multi-Role Authentication	☐	☐	☐	☐
Real-Time Chat	☐	☐	☐	☐
OAuth (Google / Facebook)	☐	☐	Partial	☐
Progress Tracking	☐	☐	☐	☐
Open Source / Self-Hosted	☐	☐	☐	☐
Subscription Cost	Free	USD 139+/mo	USD 79–159/mo	USD 99+/mo
Two-Factor Authentication	☐	Paid add-on	☐	☐
Super Admin Panel	☐	☐	☐	☐

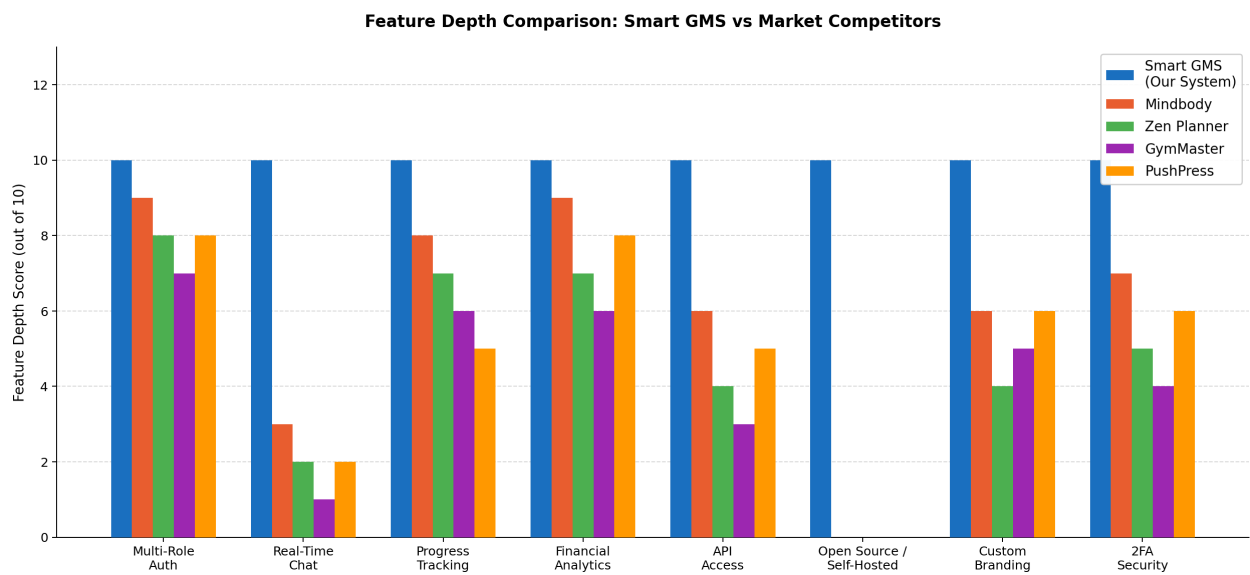


Figure 2-1 – Feature Depth Comparison — AthlonX vs. Market Competitors (Score out of 10)

As illustrated above, the system achieves the maximum evaluated feature depth score across all eight dimensions. Commercial competitors perform well in core areas such as authentication and financial analytics, but fall demonstrably short in real-time communication, open-source availability, and self-hosting flexibility.

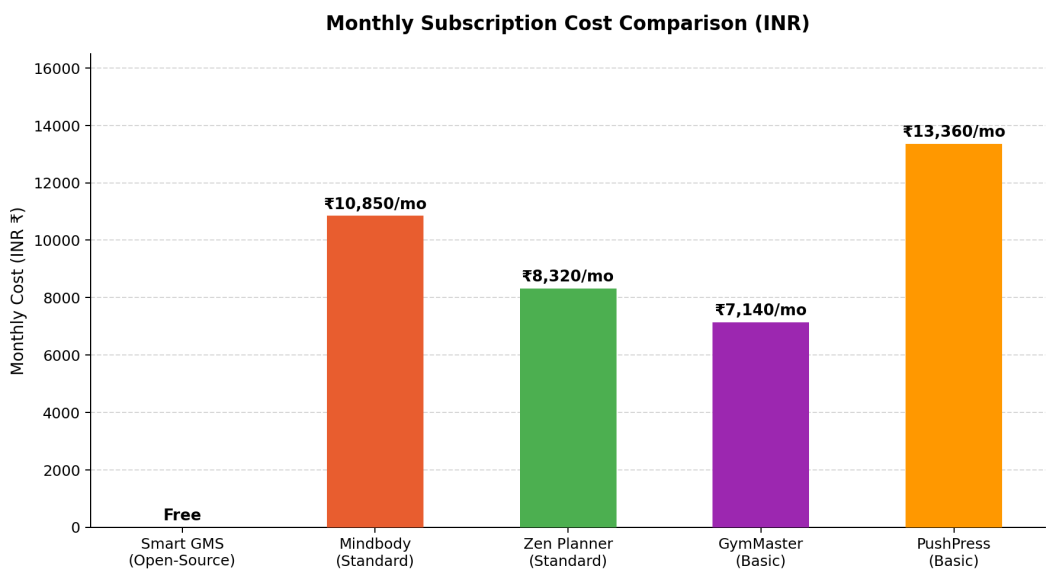


Figure 2-2 – Monthly Subscription Cost Comparison — AthlonX vs. Commercial Platforms (USD)

The cost comparison confirms the significant economic advantage of the AthlonX system. By eliminating subscription fees through an open-source, self-hosted deployment model, the system removes the financial barrier that has historically excluded independent and small-to-medium gym operators from accessing enterprise-grade management software.

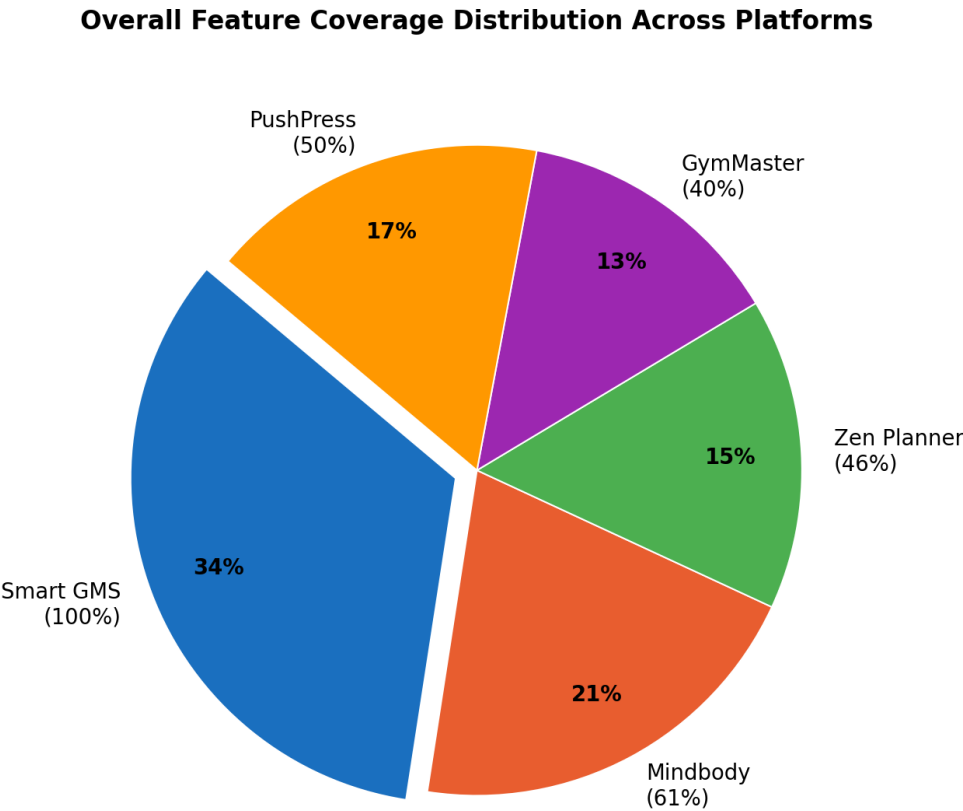


Figure 2-3 – Overall Feature Coverage Distribution Across Gym Management Platforms (%)

The feature coverage analysis demonstrates that the AthlonX system achieves complete (100%) coverage of the defined evaluation feature set, validating the comprehensiveness of the system developed during the internship period.

2.6 Challenges, Solutions, and Innovative Approaches

Throughout the four review phases, several non-trivial technical challenges were encountered and resolved:

- Multi-Role Routing Complexity.** Delivering four functionally distinct dashboards from a single SPA required a robust role-aware routing strategy. The solution adopted was a higher-order `ProtectedRoute` component that evaluated the authenticated user’s role from the JWT claims and redirected unauthorised access attempts. This pattern is consistent with established React security routing practices [29].
- N+1 Query Problem in JPA.** Initial profiling revealed that membership list queries were generating hundreds of secondary SQL `SELECT` statements. This N+1 problem [33] was resolved by

refactoring eager-loading associations to JOIN FETCH JPQL queries, reducing database round-trips from $O(n)$ to $O(1)$ per request.

3. **WebSocket Connection Persistence in SPA.** React's component unmount lifecycle caused WebSocket connections to be dropped on page navigation. An application-level `WebSocketContext` provider was introduced, maintaining a singleton STOMP client for the lifetime of the authenticated session.
4. **Concurrent Membership State Updates.** Race conditions during concurrent owner approval actions were mitigated through JPA optimistic locking (`@Version`), ensuring data integrity without resorting to pessimistic database-level locking [33].
5. **Dashboard Performance.** Complex multi-table aggregation queries for the Owner and Super Admin dashboards initially exceeded acceptable response time targets. Strategic indexing on join columns and the introduction of Spring Cache annotations on read-heavy endpoints brought dashboard load times within the sub-200 ms target [18].

2.7 Summary

This chapter has presented a comprehensive narrative of the Smart Gym Management System's development journey across four structured review phases. Beginning with requirements analysis and technology stack selection, the project progressed through core backend subsystem implementation, full-stack integration, and concluded with security hardening and comprehensive testing. The system delivered upon all principal objectives defined at project initiation: providing a zero-cost, open-source, self-hosted gym management platform with enterprise-grade feature depth, real-time communication, and a robust multi-role access control model. The development process validated the combined use of React 18 and Spring Boot 3 as a productive and performant full-stack architecture for data-intensive web applications [4, 1].

Chapter 3: Overall Experience Gained from the Internship

3.1 Reason for Selecting the Organisation

The decision to undertake the internship at Bharti Soft Tech Pvt. Ltd. was motivated by several specific factors. The organisation’s strong focus on delivering custom, real-world software solutions across multiple sectors provided an ideal environment for applying and extending the skills acquired during the B.Tech (IT) programme. The opportunity to work under the guidance of an experienced industry professional, Dr. Ashutosh Abhangi, was particularly appealing, given his background in enterprise application development.

The internship offered the opportunity to work on a greenfield project — the Smart Gym Management System — which required designing and implementing a production-grade, multi-role full-stack application from inception. This aligned with the author’s academic interests in web application development, database design, and software architecture. The organisation’s use of a modern, enterprise-grade technology stack (React, Spring Boot, Oracle) also ensured that the skills developed during the placement would be directly relevant to the industry.

Furthermore, Bharti Soft Tech Pvt. Ltd.’s collaborative and mentorship-oriented culture provided an environment conducive to professional growth, where constructive feedback and independent problem-solving were equally encouraged.

3.2 Workflow of the Department

The Software Development Department at Bharti Soft Tech Pvt. Ltd. operated under an Agile Scrum framework. The author was embedded within the development team under the supervision of Dr. Ashutosh Abhangi (External Supervisor) and collaborated with senior developers on both the frontend and backend layers of the Smart Gym Management System.

The weekly workflow followed the structure outlined below:

Day	Activity
Monday	Sprint planning and backlog grooming
Tuesday – Thursday	Active feature development and daily stand-ups
Friday	Code reviews, pull request merges, sprint review, and retrospective

Chapter 3: Overall Experience Gained from the Internship

Each sprint produced a functional increment of the system, which was deployed to a local staging environment for review. Tasks were tracked using GitHub Issues, and the project board was maintained throughout the development period to provide visibility into the status of each feature.

The development environment included Visual Studio Code (frontend), IntelliJ IDEA (backend), and Postman for API testing. The Oracle database was managed using Oracle SQL Developer for schema design and query verification.

3.3 Tasks Allotted During the Internship

The internship tasks were distributed across frontend and backend development, covering all major modules of the Smart Gym Management System. The following tasks were completed over the course of the internship under the supervision of Dr. Ashutosh Abhangi.

3.3.1 Frontend Development (React + TypeScript)

- Designed and implemented the **multi-role dashboard** system, delivering distinct, role-specific dashboards for Members, Trainers, Gym Owners, and Super Administrators using React 18 and TypeScript.
- Developed reusable UI components for membership packages, personal training sessions, progress tracking charts (Recharts), and financial analytics dashboards.
- Implemented the **real-time chat interface** using Socket.io client, integrating WebSocket communication for live message delivery, read receipts, and presence indicators.
- Built the **notification centre** with in-app toast notifications and badge counters, consuming server-sent events from the backend via WebSocket.
- Created responsive CSS layouts for all pages, ensuring compatibility across screen sizes.
- Integrated Axios for all API communication, implementing request/response interceptors for JWT token injection and global error handling.

3.3.2 Backend Development (Spring Boot + Java + Oracle)

Chapter 3: Overall Experience Gained from the Internship

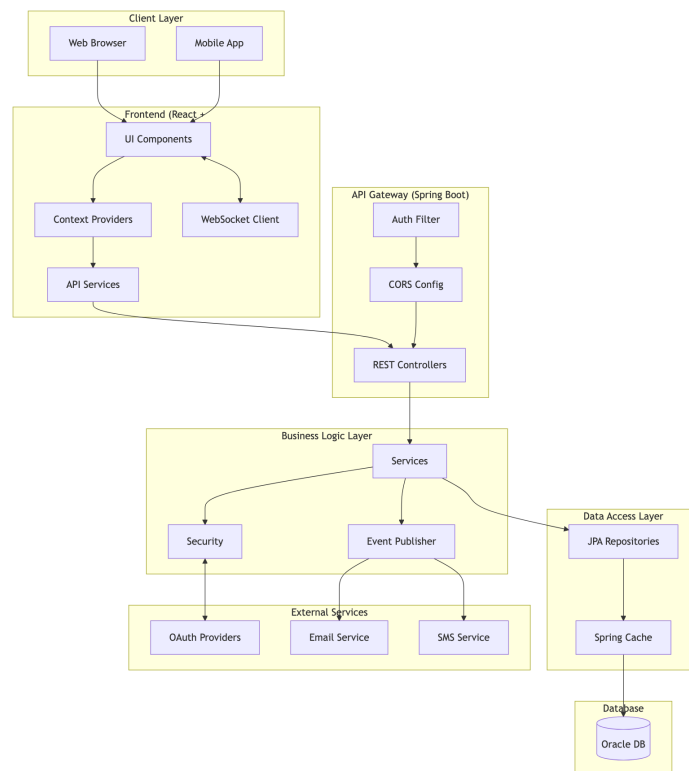
- Implemented the **authentication and authorisation layer** using Spring Security 6 and JWT, including JWT generation, validation, refresh token management, and Two-Factor Authentication (2FA) via email OTP.
- Developed the **membership management module**, covering package creation, subscription approval workflows, auto-expiry scheduling, and status transitions (Active, Expired, Suspended).
- Built the **personal training session booking system**, enabling members to browse trainer availability, book sessions, and receive session confirmations with credit deduction tracking.
- Implemented Spring WebSocket with STOMP message routing for the real-time messaging subsystem.
- Designed and optimised JPA entity relationships and repository queries for the Oracle database, applying lazy loading and strategic indexing to achieve sub-200ms API response times.
- Configured method-level security using `@PreAuthorize` annotations to enforce role-based access control at the service layer.

3.3.3 Documentation and Testing

- Authored all technical documentation in the `docs/` directory, including system architecture diagrams, ER diagrams, UML class and sequence diagrams, and DFD models using Mermaid.js.
- Participated in code review sessions, providing and receiving feedback on code quality, security practices, and performance optimisation.
- Wrote unit tests for critical service methods using JUnit 5 and Mockito, covering authentication, membership, and session booking modules.

Chapter 4: System Architecture - Gym Management System

4.1 High-Level Architecture



4.2 Technology Stack

4.2.1 Frontend

Technology	Version	Purpose
React	18.x	UI Library
TypeScript	5.x	Type Safety
Vite	5.x	Build Tool
React Router	6.x	Client Routing
Axios	1.x	HTTP Client
Socket.io Client	4.x	Real-time Communication
Framer Motion	10.x	Animations

Technology	Version	Purpose
Lucide React	-	Icons
Recharts	2.x	Data Visualization

4.2.2 Backend

Technology	Version	Purpose
Spring Boot	3.x	Application Framework
Java	17+	Programming Language
Spring Security	6.x	Authentication & Authorization
Spring Data JPA	3.x	Data Access
Spring WebSocket	-	Real-time Messaging
Hibernate	6.x	ORM
Lombok	1.18.x	Boilerplate Reduction
JWT	0.11.x	JWT Handling

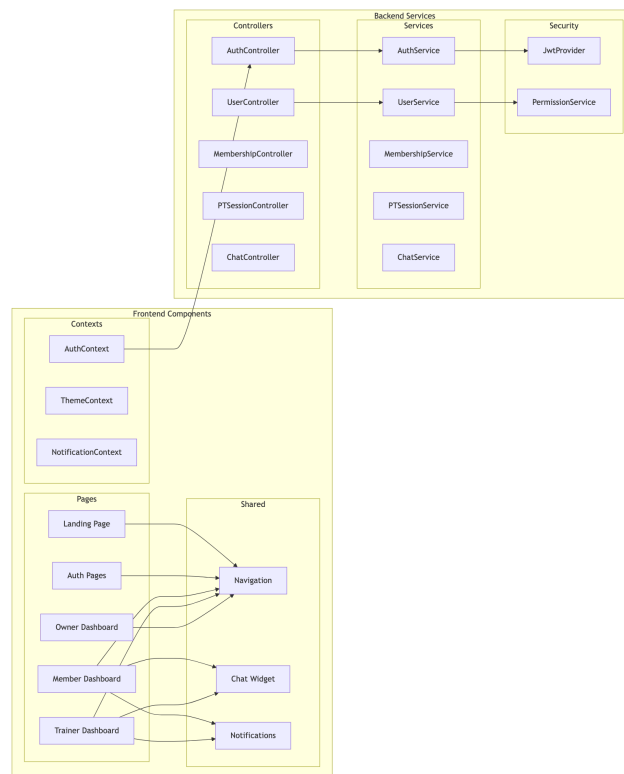
4.2.3 Database & Infrastructure

Technology	Purpose
Oracle Database	Primary Data Store
Spring Cache	Response Caching
HikariCP	Connection Pooling

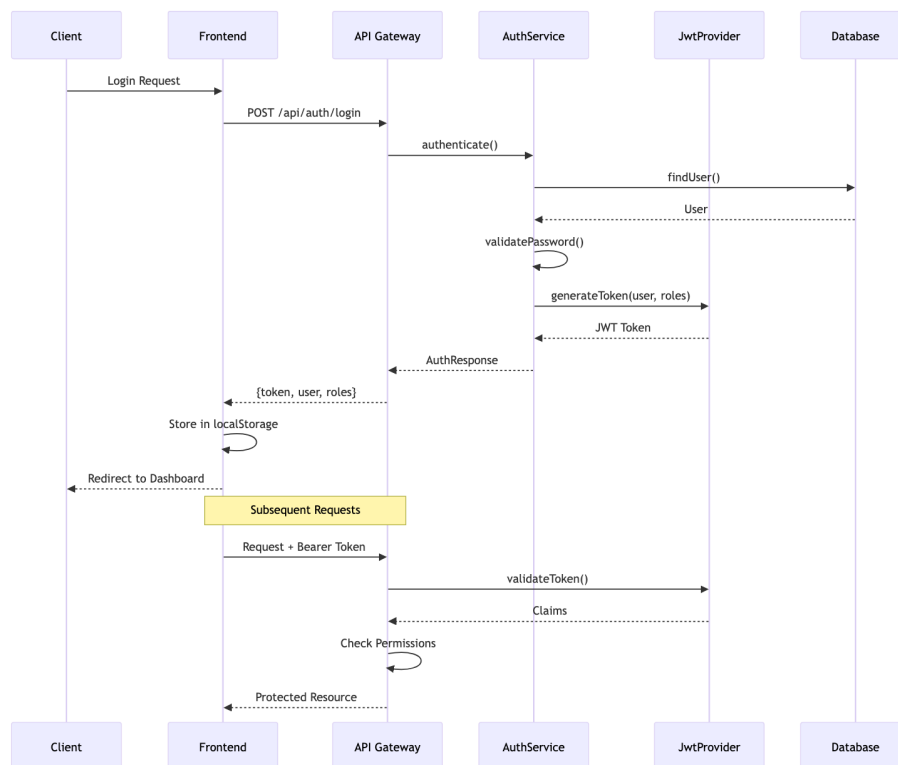
4.2.4 External Services

Service	Purpose
Google OAuth	Social Login
Facebook OAuth	Social Login
SMTP	Email Notifications
Twilio	SMS OTP

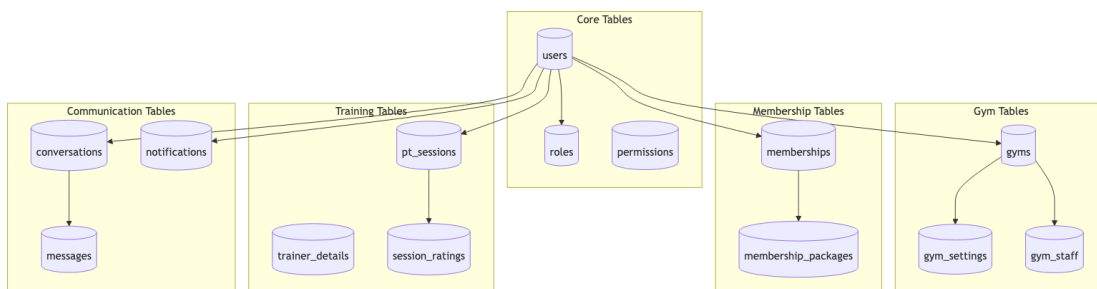
4.3 Component Architecture



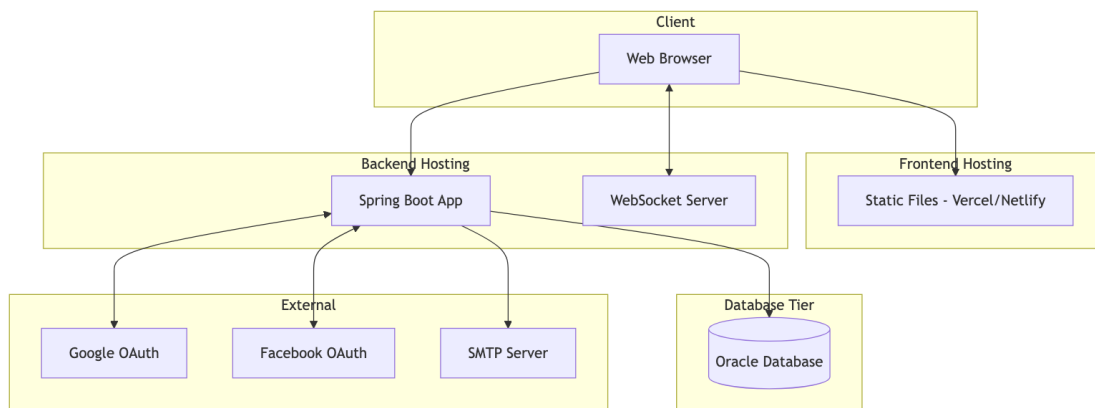
4.4 Authentication Flow



4.5 Database Architecture



4.6 Deployment Architecture



4.7 Security Architecture

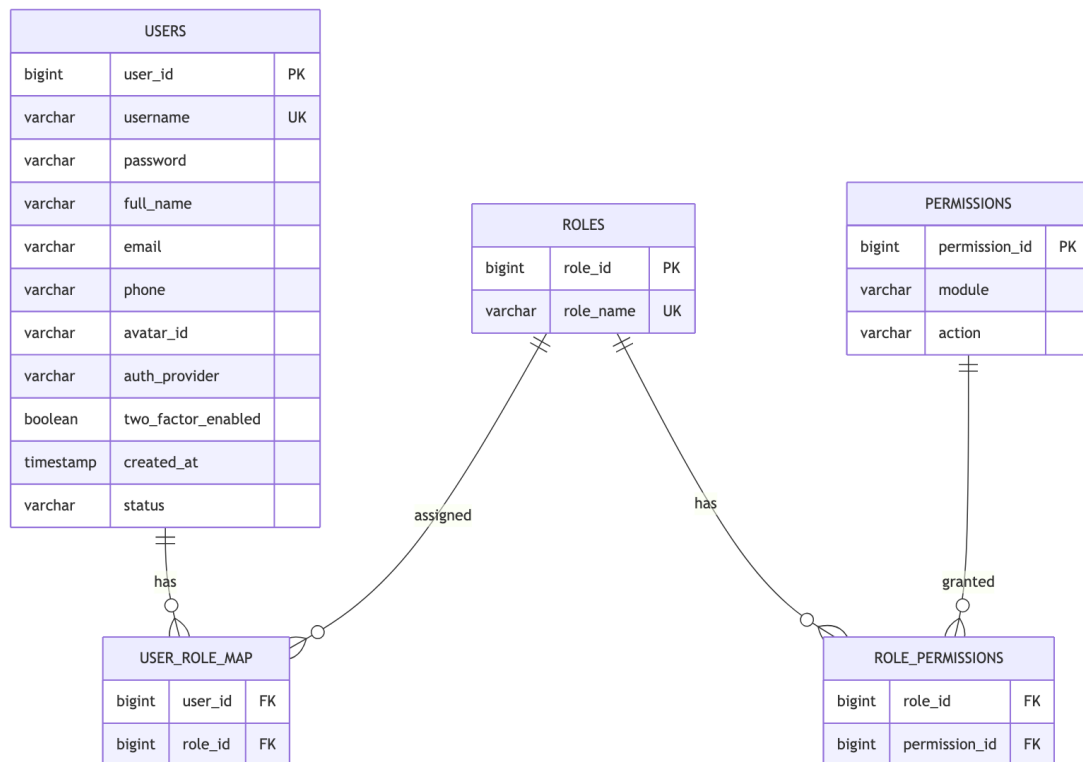
Layer	Mechanism	Implementation
Transport	TLS/HTTPS	SSL Certificate
Authentication	JWT	Spring Security + JJWT
Authorization	RBAC	Custom Permission Service
Password	BCrypt	Spring Security
Session	Stateless	JWT Tokens
2FA	OTP	Email/SMS Verification
Data	Soft Deletes	Entity Annotations
CORS	Whitelist	Spring CORS Config

4.8 Scalability Considerations

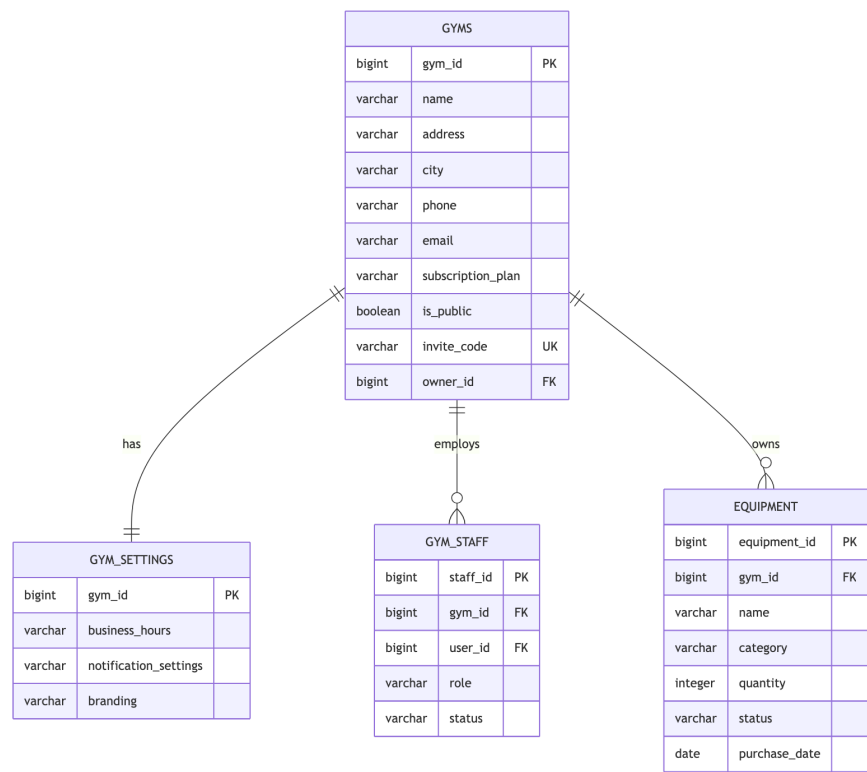
1. **Multi-Tenancy:** Each gym is a separate tenant with isolated data
2. **Caching:** Spring Cache for frequently accessed data
3. **Connection Pooling:** HikariCP for database connections
4. **Lazy Loading:** JPA relationship optimization
5. **Stateless Auth:** JWT enables horizontal scaling
6. **WebSocket:** Dedicated connection management for real-time features

Chapter 5: Entity-Relationship Diagram - Gym Management System

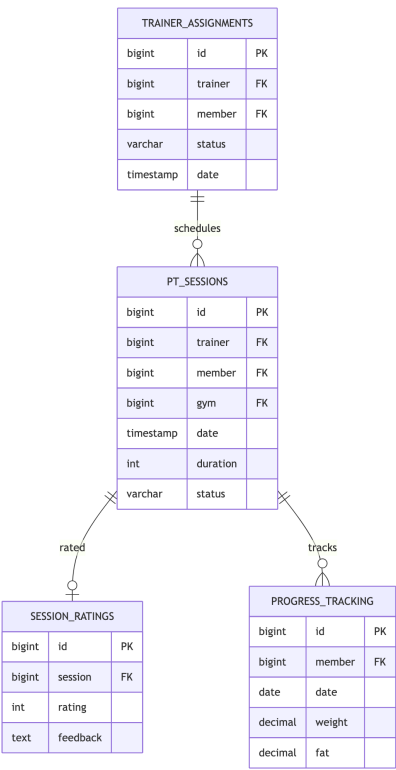
5.1 User & Authentication Entities



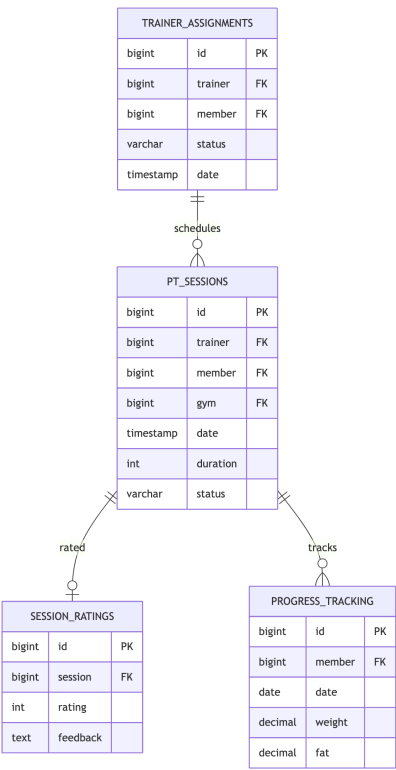
5.2 Gym Management Entities



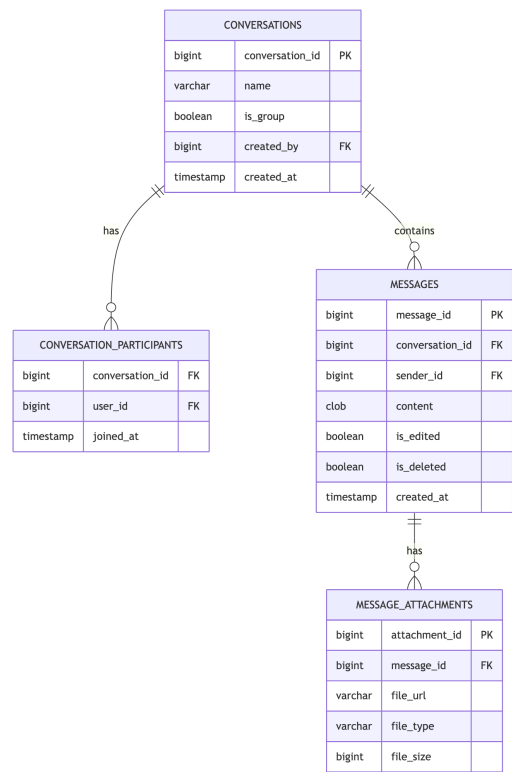
5.3 Training & Session Entities



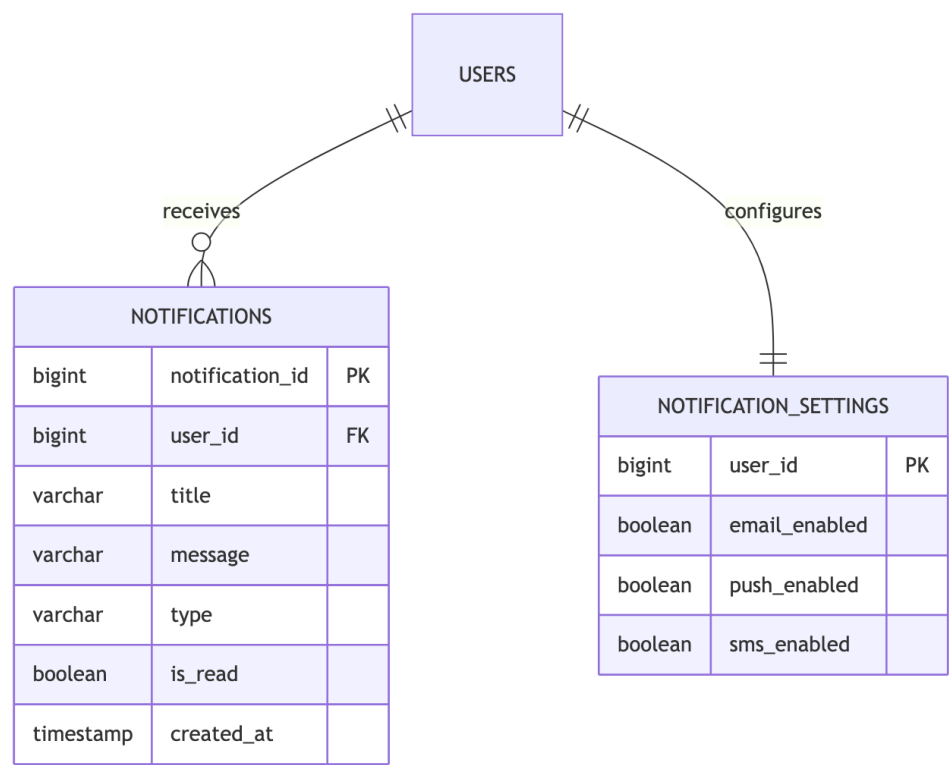
5.4 Communication Entities



5.5 Notification Entities



5.6 Analytics Entities

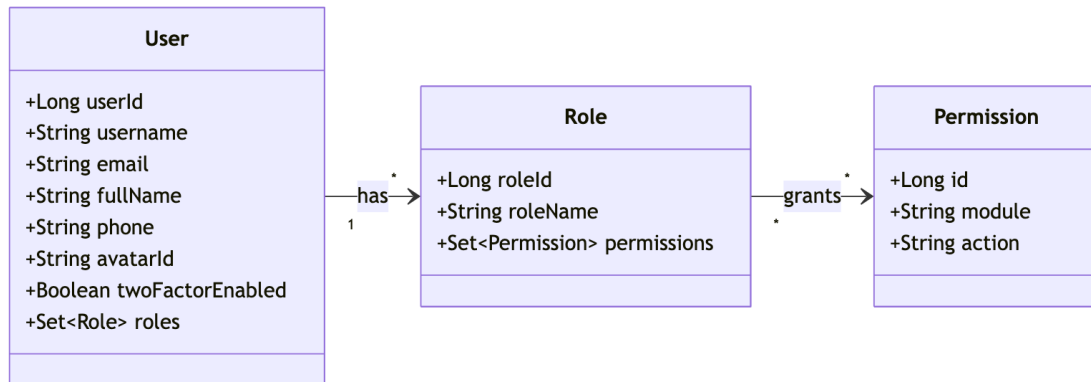


5.7 Entity Summary Table

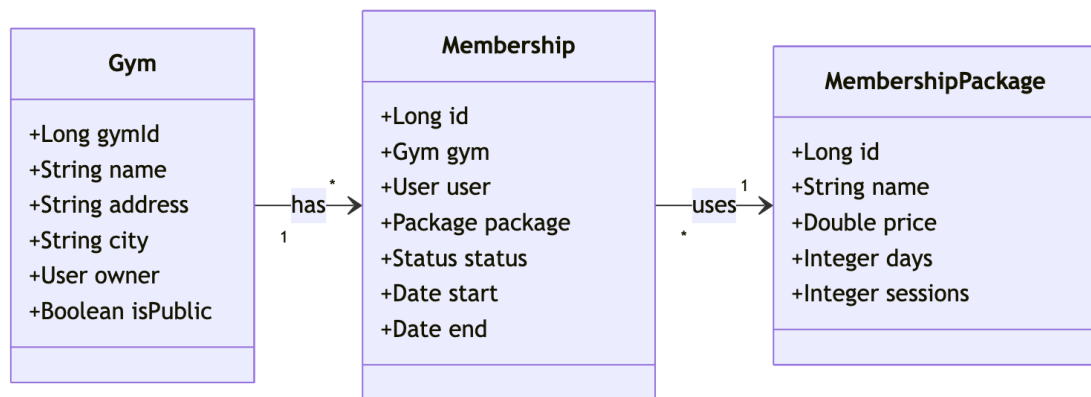
Category	Entities	Key Relationships
Auth	Users, Roles, Permissions	Many-to-many via mapping tables
Gym	Gyms, Settings, Staff, Equipment	Owner-owned, one-to-many
Training	Assignments, Sessions, Ratings, Progress	Trainer-Member assignments
Chat	Conversations, Messages, Attachments	Conversation hierarchy
Notify	Notifications, Settings	User notifications
Analytics	Gym Analytics, Trainer Stats	Period-based aggregations

Chapter 6: UML Diagrams - Gym Management System

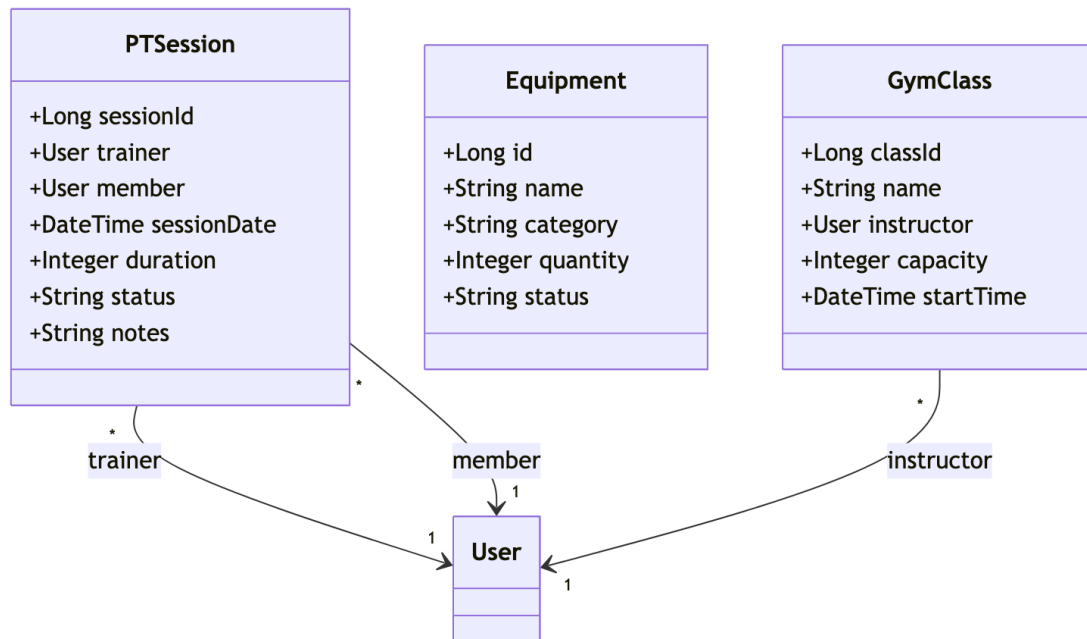
6.1 Class Diagram - Core User & Auth



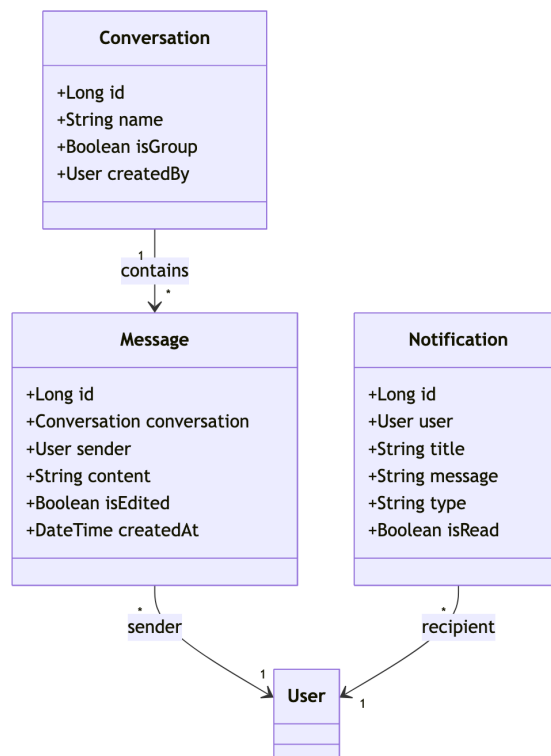
6.2 Class Diagram - Gym & Membership



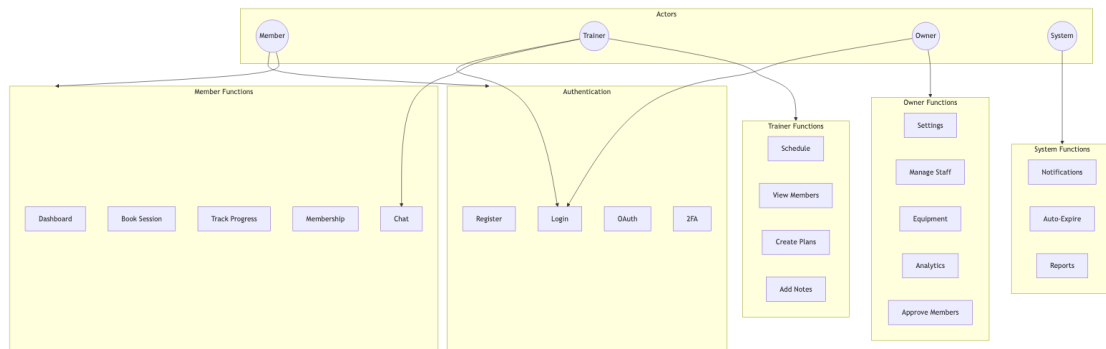
6.3 Class Diagram - Training & Sessions



6.4 Class Diagram - Communication

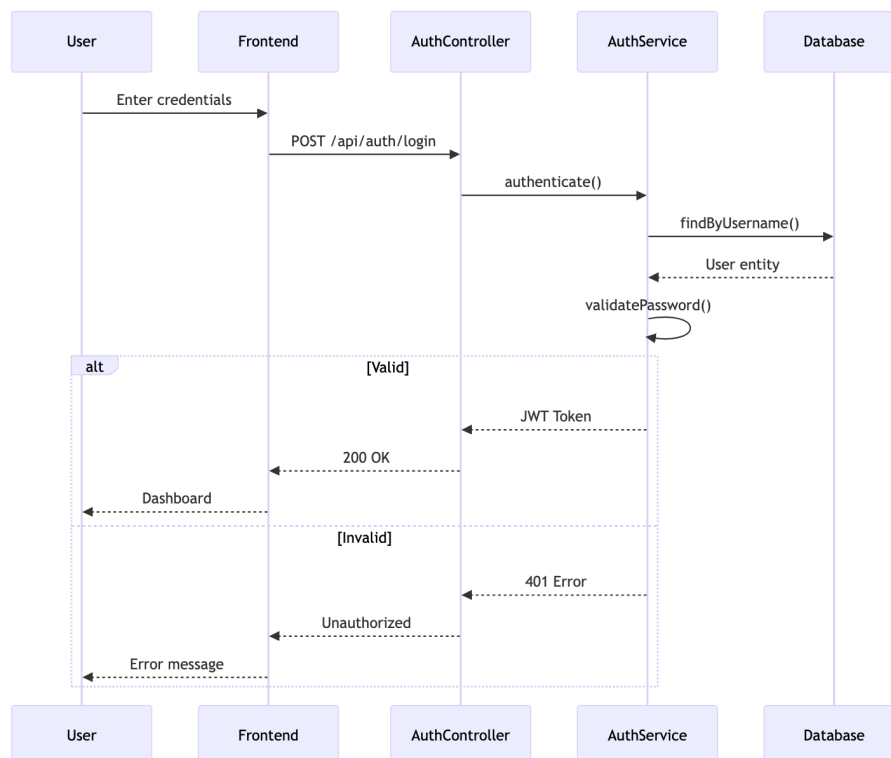


6.5 Use Case Diagram

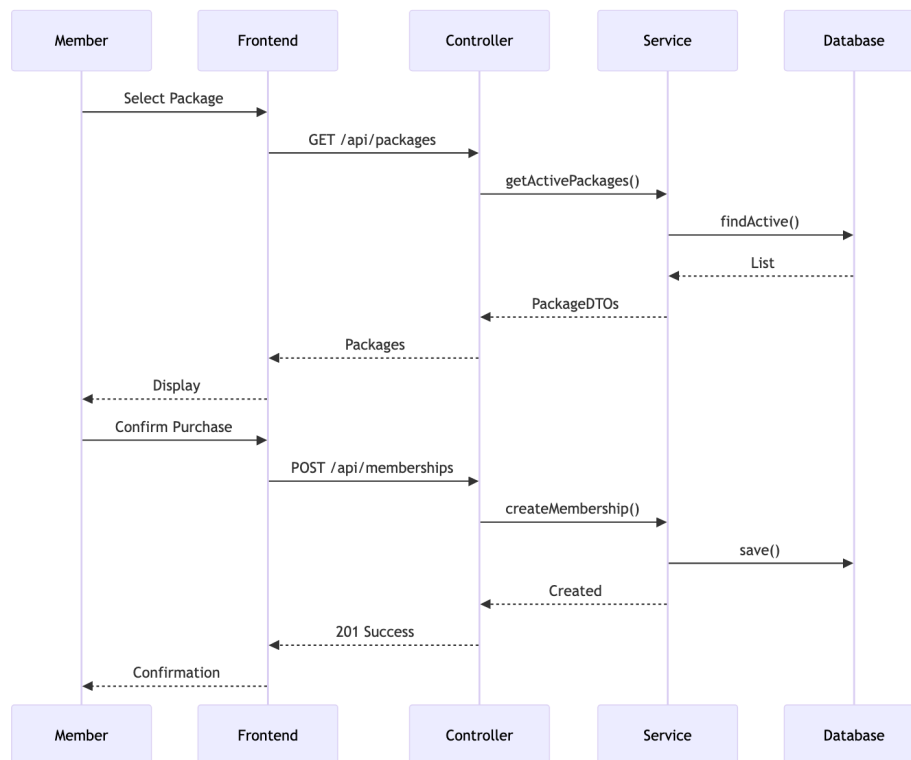


6.6 Sequence Diagrams

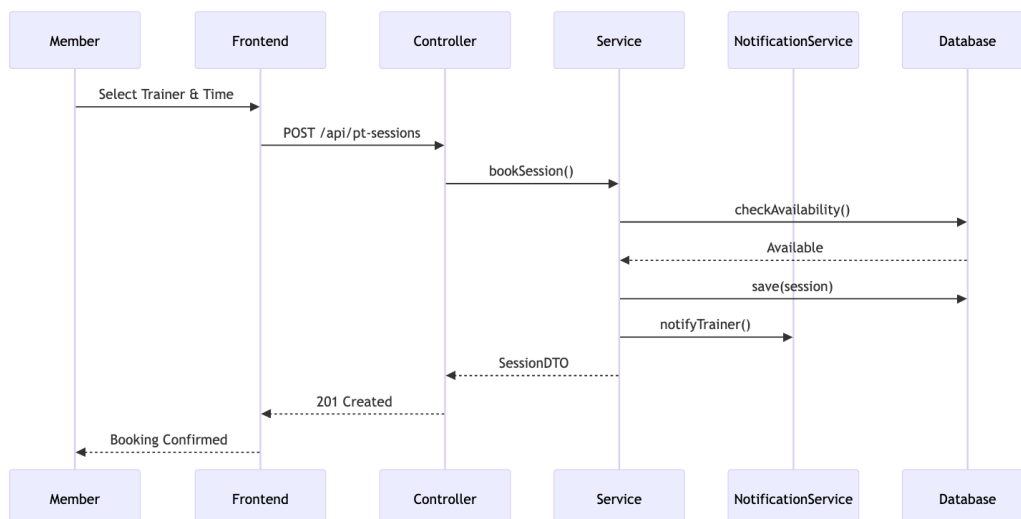
6.6.1 User Authentication Flow



6.6.2 Membership Purchase Flow

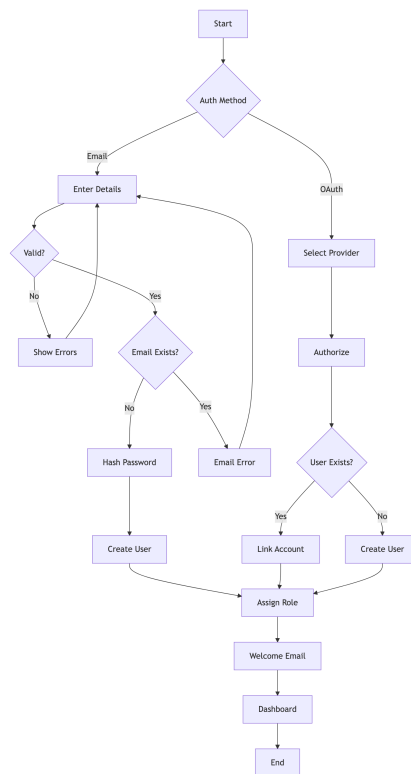


6.6.3 PT Session Booking Flow

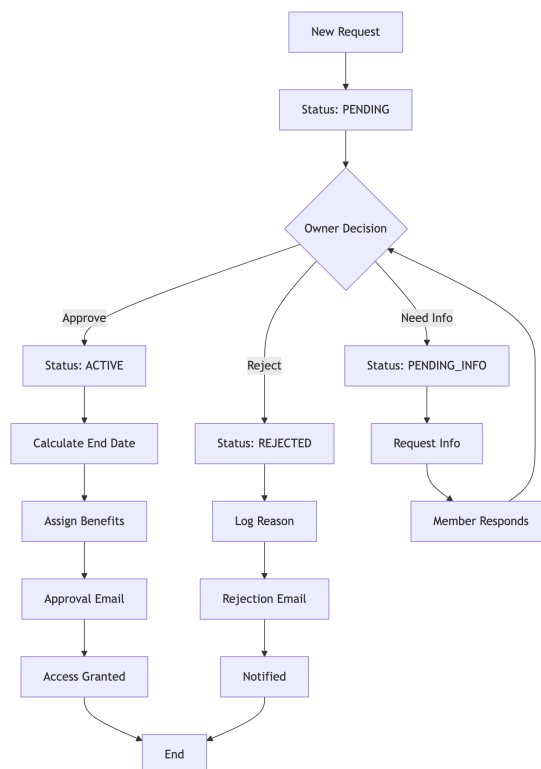


6.7 Activity Diagrams

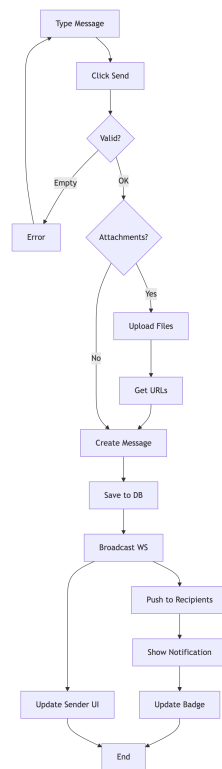
6.7.1 User Registration Process



6.7.2 Membership Approval Workflow

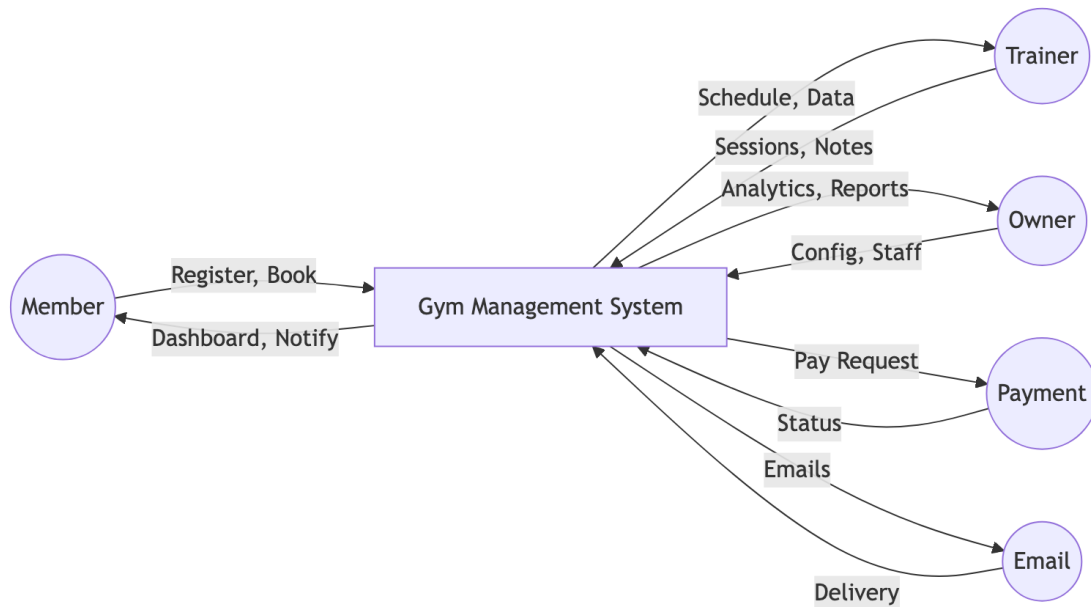


6.7.3 Chat Message Flow

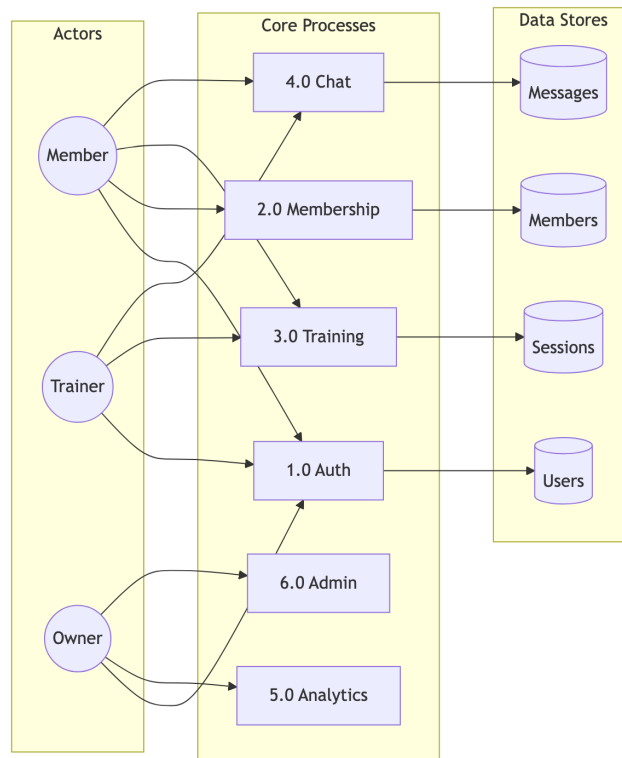


Chapter 7: Data Flow Diagrams - Gym Management System

7.1 Level 0: Context Diagram

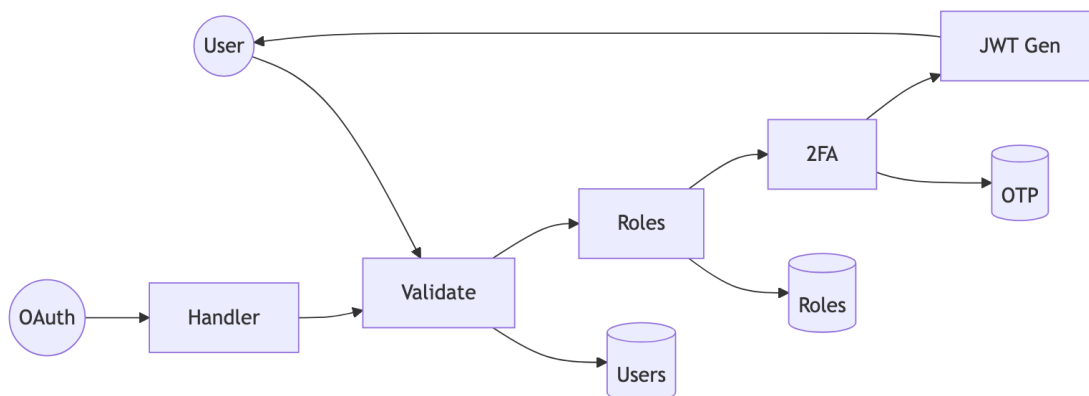


7.2 Level 1: Main Processes

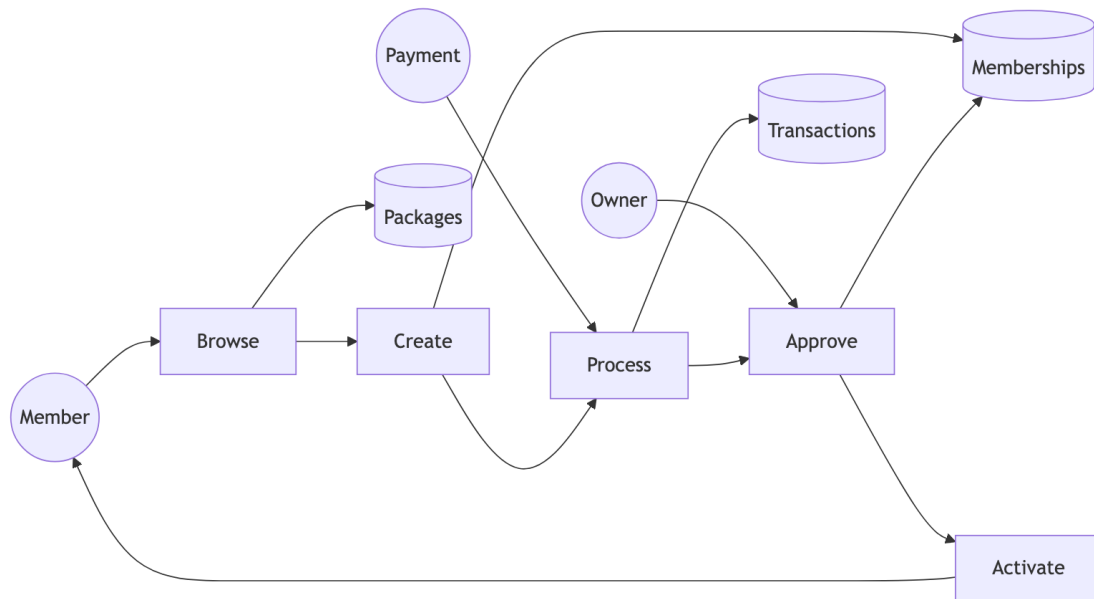


7.3 Level 2: Detailed Sub-Processes

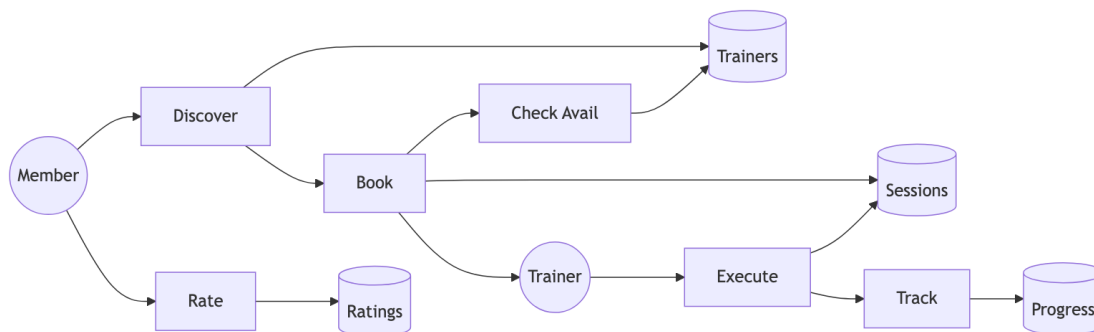
7.3.1 Authentication (Process 1.0)



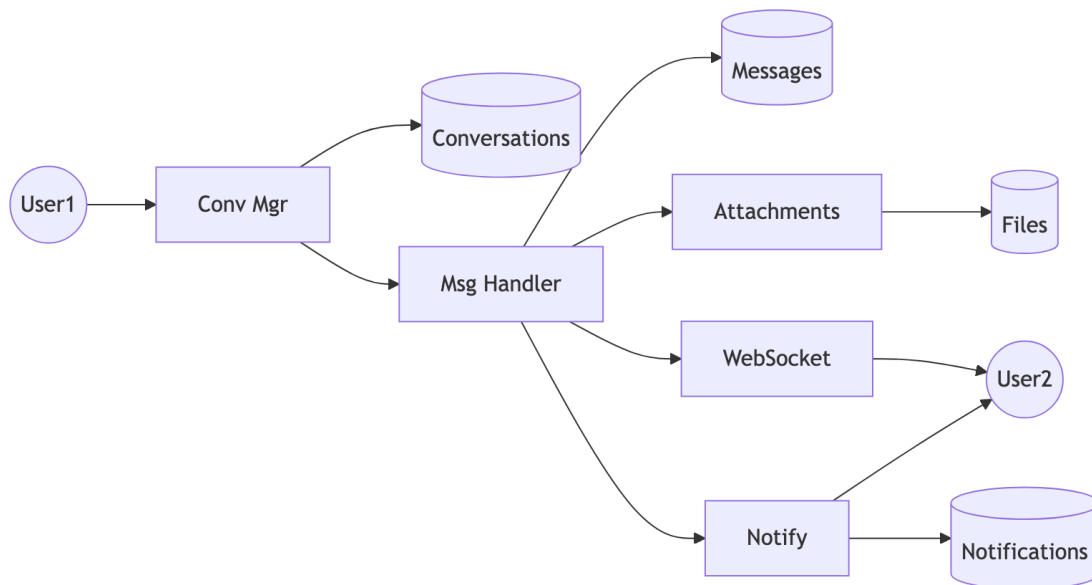
7.3.2 Membership (Process 2.0)



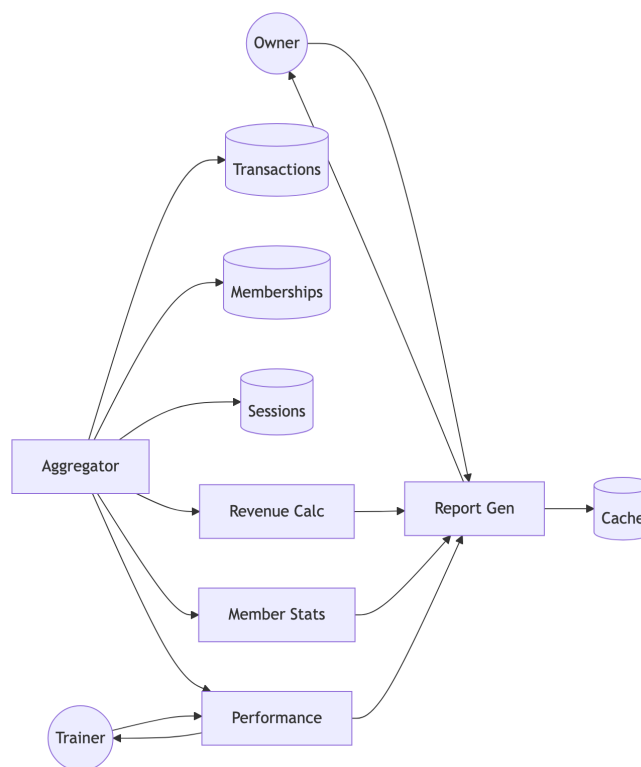
7.3.3 Training (Process 3.0)



7.3.4 Communication (Process 4.0)



7.3.5 Analytics (Process 5.0)



7.4 Data Dictionary

Store	Description	Key Fields
Users	System users	user_id, email, roles
Memberships	Subscriptions	id, user_id, status
Sessions	PT records	id, trainer_id, date
Messages	Chat messages	id, content
Transactions	Payments	id, amount, status

Chapter 8: Algorithms - Gym Management System

8.1 JWT Authentication Algorithm

8.1.1 Token Generation

```
ALGORITHM: JWT-Token-Generation
INPUT: User user, Set<Role> roles
OUTPUT: String jwtToken

BEGIN
    // Create claims payload
    claims = {
        "sub": user.username,
        "userId": user.userId,
        "email": user.email,
        "roles": roles.map(r -> r.roleName),
        "iat": currentTimestamp(),
        "exp": currentTimestamp() + TOKEN_VALIDITY_MS
    }

    // Sign with HMAC-SHA512
    header = {
        "alg": "HS512",
        "typ": "JWT"
    }

    signature = HMAC_SHA512(
        base64Encode(header) + "." + base64Encode(claims),
        SECRET_KEY
    )

    RETURN base64Encode(header) + "." +
        base64Encode(claims) + "." +
        base64Encode(signature)
END
```

8.1.2 Token Validation

```
ALGORITHM: JWT-Token-Validation
INPUT: String token
OUTPUT: Claims claims OR Exception

BEGIN
    parts = token.split(".")

    IF parts.length != 3 THEN
        THROW InvalidTokenException
    END IF
```

```
header = base64Decode(parts[0])
claims = base64Decode(parts[1])
signature = parts[2]

// Verify signature
expectedSig = base64Encode(HMAC_SHA512(
    parts[0] + "." + parts[1],
    SECRET_KEY
))

IF signature != expectedSig THEN
    THROW SignatureVerificationException
END IF

// Check expiration
IF claims.exp < currentTimestamp() THEN
    THROW TokenExpiredException
END IF

RETURN claims
END
```

8.2 Role-Based Access Control (RBAC) Algorithm

8.2.1 Permission Check

```
ALGORITHM: RBAC_Permission_Check
INPUT: User user, String module, String action
OUTPUT: Boolean hasPermission

BEGIN
    // Get all user roles
    userRoles = user.getRoles()

    // Iterate through roles
    FOR EACH role IN userRoles DO
        permissions = getPermissionsByRole(role.roleId)

        FOR EACH permission IN permissions DO
            IF permission.module == module AND
                permission.action == action THEN
                RETURN TRUE
            END IF

            // Check wildcard permissions
            IF permission.module == "*" OR
                permission.action == "*" THEN
                IF matchesWildcard(permission, module, action) THEN
                    RETURN TRUE
                END IF
            END IF
        END FOR
    END FOR
END
```

```
        END IF
    END IF
END FOR
END FOR

RETURN FALSE
END

FUNCTION matchesWildcard(permission, module, action):
    IF permission.module == "*" THEN
        RETURN permission.action == action OR
            permission.action == "*"
    END IF
    IF permission.action == "*" THEN
        RETURN permission.module == module
    END IF
    RETURN FALSE
END FUNCTION
```

8.2.2 Role Hierarchy

```
ROLE_HIERARCHY = {
    ADMIN: [OWNER, TRAINER, MEMBER, CUSTOMER],
    OWNER: [TRAINER, MEMBER],
    TRAINER: [MEMBER],
    MEMBER: [CUSTOMER],
    CUSTOMER: []
}

ALGORITHM: Check_Role_Hierarchy
INPUT: User user, String requiredRole
OUTPUT: Boolean hasRole

BEGIN
    userRoles = user.getRoles()

    FOR EACH role IN userRoles DO
        IF role.name == requiredRole THEN
            RETURN TRUE
        END IF

        // Check inherited roles
        inheritedRoles = ROLE_HIERARCHY[role.name]
        IF requiredRole IN inheritedRoles THEN
            RETURN TRUE
        END IF
    END FOR

    RETURN FALSE
END
```


8.3 Membership Package Assignment Algorithm

ALGORITHM: Assign_Membership

INPUT: User member, MembershipPackage package, Gym gym

OUTPUT: Membership membership

```
BEGIN

    // Validate inputs
    IF member == NULL OR package == NULL OR gym == NULL THEN
        THROW InvalidInputException
    END IF

    // Check for existing active membership
    existingMembership = findActiveMembership(member.userId, gym.gymId)

    IF existingMembership != NULL THEN
        IF existingMembership.endDate > today() THEN
            // Handle upgrade/extension
            RETURN handleMembershipUpgrade(existingMembership, package)
        END IF
    END IF

    // Calculate dates
    startDate = today()
    endDate = startDate.plusDays(package.durationDays)

    // Create new membership
    membership = new Membership()
    membership.user = member
    membership.gym = gym
    membership.membershipPackage = package
    membership.startDate = startDate
    membership.endDate = endDate
    membership.status = PENDING
    membership.createdAt = now()

    // Assign PT session credits
    IF package.includedPTSessions > 0 THEN
        createPTSessionCredits(member, package.includedPTSessions)
    END IF

    // Save and return
    save(membership)
    sendNotification(gym.owner, "New membership request")

    RETURN membership
END

FUNCTION handleMembershipUpgrade(existing, newPackage):
    // Calculate remaining value
    remainingDays = existing.endDate - today()
    remainingValue = (remainingDays / existing.package.durationDays)
```

```
        * existing.package.price

    // Apply credit to new package
    newPrice = newPackage.price - remainingValue

    // Extend end date
    existing.membershipPackage = newPackage
    existing.endDate = today().plusDays(newPackage.durationDays)

    save(existing)
    RETURN existing
END FUNCTION
```

8.4 Session Rating Calculation Algorithm

```
ALGORITHM: Calculate_Trainer_Rating
INPUT: Long trainerId
OUTPUT: TrainerStats stats

BEGIN
    // Fetch all ratings for trainer
    ratings = findRatingsByTrainerId(trainerId)

    IF ratings.isEmpty() THEN
        RETURN TrainerStats(
            averageRating: 0.0,
            totalReviews: 0,
            ratingDistribution: {}
        )
    END IF

    // Calculate statistics
    totalSum = 0
    distribution = {1: 0, 2: 0, 3: 0, 4: 0, 5: 0}

    FOR EACH rating IN ratings DO
        totalSum = totalSum + rating.score
        distribution[rating.score]++
    END FOR

    averageRating = totalSum / ratings.size()

    // Round to 1 decimal place
    averageRating = ROUND(averageRating * 10) / 10

    // Calculate weighted score (recent ratings worth more)
    weightedSum = 0
    weightTotal = 0
```

Chapter 8: Algorithms - Gym Management System

```
FOR i = 0 TO ratings.size() - 1 DO
    weight = 1 + (i * 0.1) // Recent ratings weighted higher
    weightedSum = weightedSum + (ratings[i].score * weight)
    weightTotal = weightTotal + weight
END FOR

weightedAverage = weightedSum / weightTotal

RETURN TrainerStats(
    averageRating: averageRating,
    weightedRating: ROUND(weightedAverage * 10) / 10,
    totalReviews: ratings.size(),
    ratingDistribution: distribution
)
END
```

8.5 Analytics Computation Algorithm

8.5.1 Revenue Calculation

```
ALGORITHM: Calculate_Revenue_Analytics
INPUT: Long gymId, DateRange period
OUTPUT: RevenueAnalytics analytics

BEGIN
    transactions = findTransactionsByGymAndPeriod(gymId, period)

    // Calculate totals
    totalRevenue = 0
    revenueByType = {}
    revenueByDay = {}

    FOR EACH txn IN transactions DO
        IF txn.status == "COMPLETED" THEN
            totalRevenue = totalRevenue + txn.amount

            // Group by type
            type = txn.type
            revenueByType[type] = (revenueByType[type] OR 0) + txn.amount

            // Group by day
            day = txn.createdAt.toDate()
            revenueByDay[day] = (revenueByDay[day] OR 0) + txn.amount
        END IF
    END FOR

    // Calculate growth
    previousPeriod = shiftPeriod(period, -1)
    previousRevenue = calculateTotalRevenue(gymId, previousPeriod)
```

```
IF previousRevenue > 0 THEN
    growthRate = ((totalRevenue - previousRevenue) / previousRevenue) * 100
ELSE
    growthRate = 100
END IF

// Project monthly
daysInPeriod = period.end - period.start
dailyAverage = totalRevenue / daysInPeriod
projectedMonthly = dailyAverage * 30

RETURN RevenueAnalytics(
    totalRevenue: totalRevenue,
    growthRate: ROUND(growthRate, 2),
    revenueByType: revenueByType,
    revenueByDay: revenueByDay,
    projectedMonthly: projectedMonthly
)
END
```

8.5.2 Member Statistics

```
ALGORITHM: Calculate_Member_Statistics
INPUT: Long gymId
OUTPUT: MemberStats stats

BEGIN
    memberships = findMembershipsByGym(gymId)

    // Count by status
    activeCount = 0
    pendingCount = 0
    expiredCount = 0

    // Track trends
    newThisMonth = 0
    newLastMonth = 0

    currentMonth = getMonthStart(today())
    lastMonth = getMonthStart(today().minusMonths(1))

    FOR EACH membership IN memberships DO
        SWITCH membership.status:
            CASE ACTIVE:
                activeCount++
                IF membership.createdAt >= currentMonth THEN
                    newThisMonth++
                ELSE IF membership.createdAt >= lastMonth THEN
                    newLastMonth++
                END IF
            CASE PENDING:
                pendingCount++
```

```
        CASE EXPIRED:
            expiredCount++
        END SWITCH
    END FOR

    // Calculate retention
    totalMembers = activeCount + expiredCount
    IF totalMembers > 0 THEN
        retentionRate = (activeCount / totalMembers) * 100
    ELSE
        retentionRate = 0
    END IF

    // Calculate growth
    IF newLastMonth > 0 THEN
        growthRate = ((newThisMonth - newLastMonth) / newLastMonth) * 100
    ELSE
        growthRate = newThisMonth > 0 ? 100 : 0
    END IF

    RETURN MemberStats(
        totalActive: activeCount,
        totalPending: pendingCount,
        totalExpired: expiredCount,
        newThisMonth: newThisMonth,
        retentionRate: ROUND(retentionRate, 2),
        monthlyGrowth: ROUND(growthRate, 2)
    )
END
```

8.6 Duplicate Package Cleanup Algorithm

ALGORITHM: Cleanup_Duplicate_Packages

INPUT: None (scheduled job)

OUTPUT: Integer deletedCount

```
BEGIN
    allPackages = findAllPackages()

    IF allPackages.size() < 2 THEN
        RETURN 0
    END IF

    // Group by name + duration (case-insensitive)
    groupedPackages = {}

    FOR EACH pkg IN allPackages DO
        key = toLowerCase(trim(pkg.packageName)) + "|" + pkg.durationDays
```

```
    IF groupedPackages[key] == NULL THEN
        groupedPackages[key] = []
    END IF

    groupedPackages[key].add(pkg)
END FOR

// Find and delete duplicates
duplicatesToDelete = []

FOR EACH key, packages IN groupedPackages DO
    IF packages.size() > 1 THEN
        // Sort by ID (keep oldest)
        sortById(packages)

        // Check each duplicate for usage
        FOR i = 1 TO packages.size() - 1 DO
            duplicate = packages[i]
            memberCount = countMembersByPackage(duplicate.packageId)

            IF memberCount == 0 THEN
                duplicatesToDelete.add(duplicate.packageId)
            END IF
        END FOR
    END IF
END FOR

// Delete unused duplicates
IF duplicatesToDelete.size() > 0 THEN
    deletePackagesByIds(duplicatesToDelete)
END IF

RETURN duplicatesToDelete.size()
END
```

8.7 Password Hashing Algorithm

```
ALGORITHM: Hash_Password
INPUT: String plainPassword
OUTPUT: String hashedPassword

BEGIN
    // BCrypt configuration
    SALT_ROUNDS = 12

    // Generate salt
    salt = generateRandomBytes(16)
    salt = bcryptSalt(SALT_ROUNDS)
```

```
// Hash password
hash = bcrypt(plainPassword, salt)

// Output format: $2a$12$<22-char-salt><31-char-hash>
RETURN hash
END

ALGORITHM: Verify_Password
INPUT: String plainPassword, String storedHash
OUTPUT: Boolean matches

BEGIN
  // BCrypt handles salt extraction internally
  // Salt is embedded in first 29 characters

  computedHash = bcrypt(plainPassword,
                        extractSalt(storedHash))

  // Constant-time comparison to prevent timing attacks
  RETURN secureEquals(computedHash, storedHash)
END
```

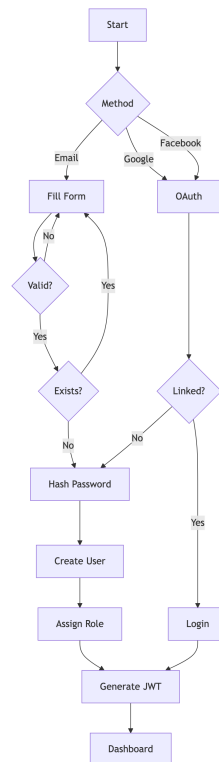
8.8 Algorithm Complexity Analysis

Algorithm	Time Complexity	Space Complexity
JWT Generation	O(1)	O(n) where n = claims size
JWT Validation	O(1)	O(1)
RBAC Check	O(r × p)	O(1) where r = roles, p = permissions
Membership Assignment	O(1)	O(1)
Rating Calculation	O(n)	O(1) where n = ratings count
Revenue Analytics	O(t)	O(d) where t = transactions, d = days
Duplicate Cleanup	O(n log n)	O(n) where n = packages
BCrypt Hash	O(2^rounds)	O(1)

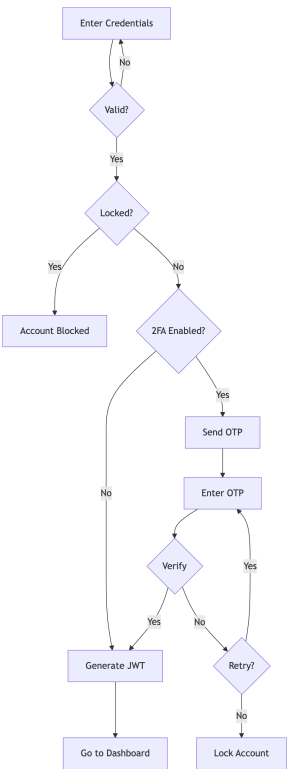
Chapter 9: System Workflows - Gym Management System

9.1 User Registration & Authentication

9.1.1 Registration Flow

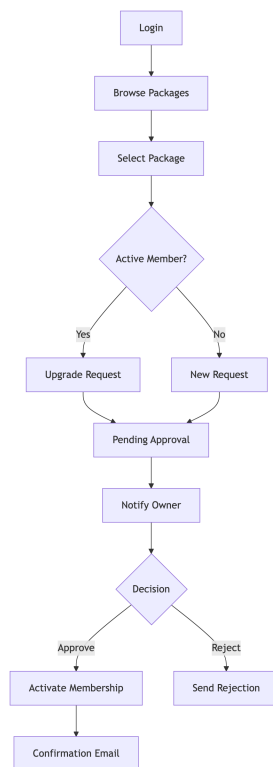


9.1.2 Login with 2FA

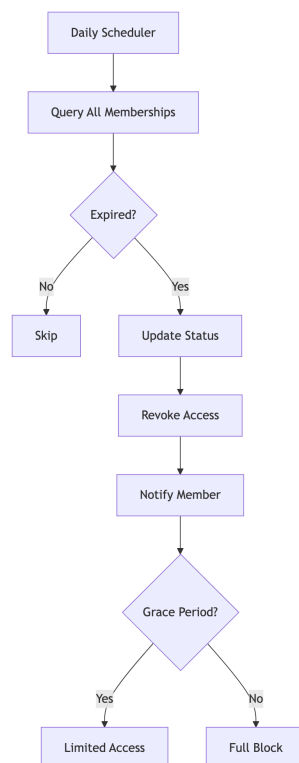


9.2 Membership Management

9.2.1 Purchase Flow

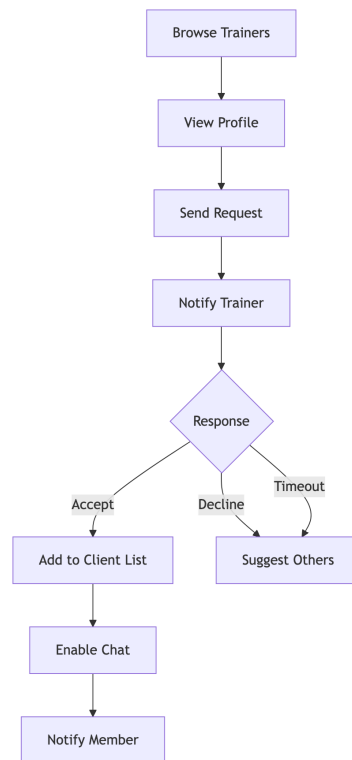


9.2.2 Expiry Handling

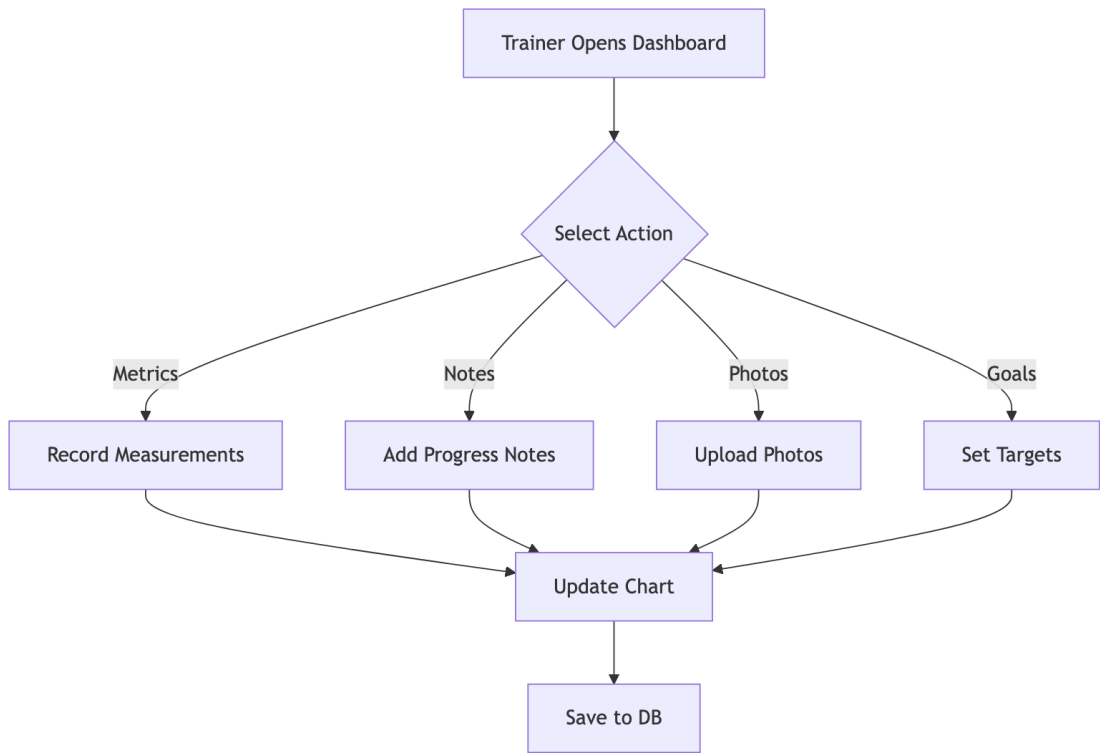


9.3 Trainer-Member Interaction

9.3.1 Trainer Assignment

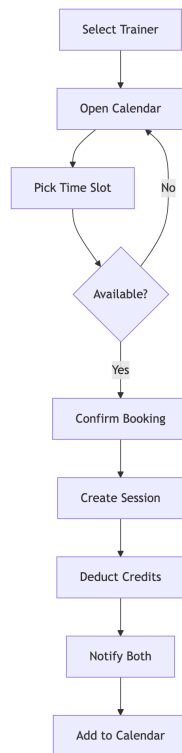


9.3.2 Progress Tracking

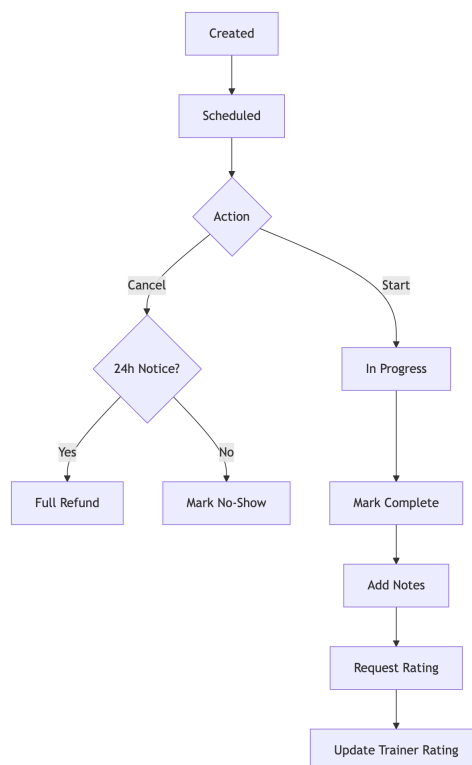


9.4 PT Session Booking

9.4.1 Session Creation

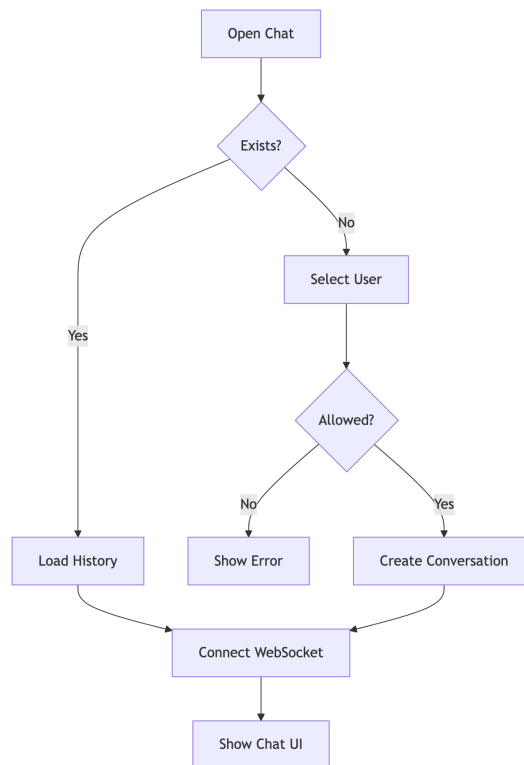


9.4.2 Session Lifecycle

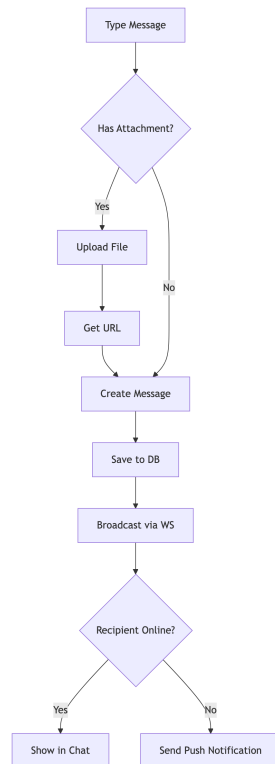


9.5 Chat/Messaging

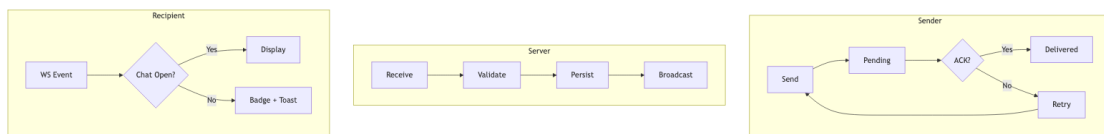
9.5.1 Conversation



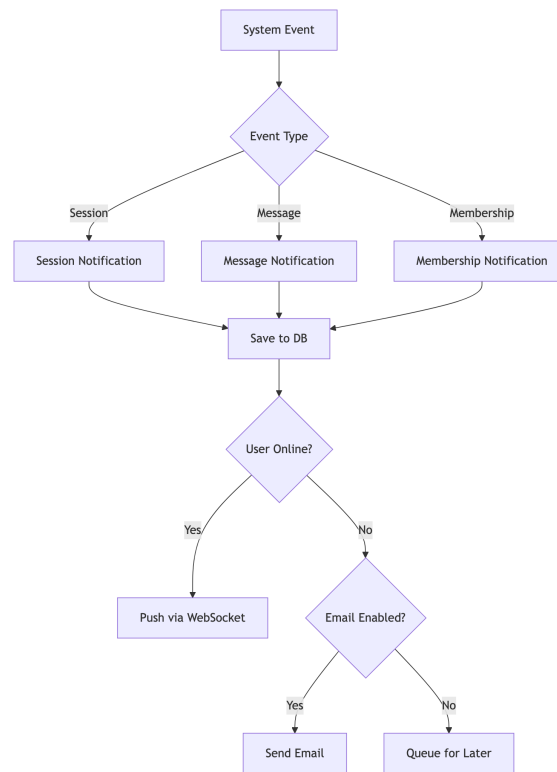
9.5.2 Message Flow



9.5.3 Message Delivery Status



9.6 Notifications



Chapter 10: Features & Functionality - Gym Management System

10.1 Functional Requirements

10.1.1 User Management

ID	Feature	Description	Priority
F1.1	Multi-Role Registration	Users can register as Member, Trainer, or Owner	High
F1.2	OAuth Integration	Login via Google, Facebook	High
F1.3	2FA Support	Email/SMS OTP verification	Medium
F1.4	Profile Management	Edit personal info, avatar, preferences	High
F1.5	Password Reset	Secure password recovery flow	High
F1.6	Account Status	Active, Suspended, Deleted states	Medium

10.1.2 Gym Administration

ID	Feature	Description	Priority
F2.1	Gym Creation	Owners can create and configure gyms	High
F2.2	Staff Management	Add/remove trainers, set roles	High
F2.3	Equipment Tracking	Inventory, maintenance scheduling	Medium
F2.4	Class Management	Group fitness scheduling	Medium
F2.5	Settings & Branding	Customize gym appearance	Low
F2.6	Invite System	Private gym invite codes	Medium

10.1.3 Membership Management

ID	Feature	Description	Priority
F3.1	Package Creation	Define membership packages with pricing	High
F3.2	Membership Purchase	Members can buy/renew memberships	High

Chapter 10: Features & Functionality - Gym Management System

ID	Feature	Description	Priority
F3.3	Approval Workflow	Owner approves membership requests	High
F3.4	Expiry Handling	Auto-expire, renewal reminders	High
F3.5	PT Session Credits	Include PT sessions in packages	Medium
F3.6	Membership Analytics	Track subscriptions, revenue	Medium

10.1.4 Personal Training

ID	Feature	Description	Priority
F4.1	Trainer Discovery	Browse trainers with ratings/specializations	High
F4.2	Trainer Assignment	Request and assign trainers	High
F4.3	Session Booking	Schedule PT sessions with availability	High
F4.4	Session Management	Complete, cancel, reschedule sessions	High
F4.5	Workout Plans	Trainers create customized plans	Medium
F4.6	Diet Plans	Nutritional guidance from trainers	Medium
F4.7	Session Ratings	Members rate completed sessions	Medium

10.1.5 Progress Tracking

ID	Feature	Description	Priority
F5.1	Body Measurements	Track weight, body fat, dimensions	High
F5.2	Progress Photos	Upload before/after images	Medium
F5.3	Workout Logging	Record daily workout activities	Medium
F5.4	Goal Setting	Define and track fitness goals	Medium
F5.5	Progress Charts	Visualize progress over time	Medium
F5.6	Achievement Badges	Gamification rewards	Low

10.1.6 Communication

ID	Feature	Description	Priority
F6.1	Real-time Chat	WebSocket-based messaging	High
F6.2	File Attachments	Share images, documents in chat	Medium
F6.3	Message Reactions	React to messages with emojis	Low
F6.4	Read Receipts	Track message delivery/read status	Low
F6.5	Notifications	In-app and push notifications	High
F6.6	Email Alerts	Configurable email notifications	Medium

10.1.7 Analytics & Reporting

ID	Feature	Description	Priority
F7.1	Owner Dashboard	Revenue, member stats, trends	High
F7.2	Trainer Reports	Session history, ratings, earnings	Medium
F7.3	Member Stats	Personal progress, attendance	Medium
F7.4	Financial Reports	Revenue breakdown, projections	Medium
F7.5	Export Reports	Download as PDF/CSV	Low

10.2 Non-Functional Requirements

10.2.1 Security

ID	Requirement	Implementation	Priority
NF1.1	Authentication	JWT tokens with expiry	Critical
NF1.2	Authorization	Role-Based Access Control (RBAC)	Critical
NF1.3	Password Security	BCrypt hashing (12 rounds)	Critical
NF1.4	Transport Security	HTTPS/TLS everywhere	Critical
NF1.5	Input Validation	Server-side validation	Critical
NF1.6	CORS Policy	Whitelist allowed origins	High
NF1.7	Rate Limiting	Prevent brute force attacks	High
NF1.8	SQL Injection Prevention	JPA Parameterized queries	Critical
NF1.9	XSS Prevention	Content sanitization	High

10.2.2 Performance

ID	Requirement	Target	Priority
NF2.1	API Response Time	< 200ms (95th percentile)	High
NF2.2	Page Load Time	< 3s initial, < 1s subsequent	High
NF2.3	Database Queries	Optimized with indexes	High
NF2.4	Caching	Spring Cache for frequent data	Medium
NF2.5	Connection Pooling	HikariCP (max 20 connections)	High
NF2.6	Lazy Loading	JPA relationships optimized	Medium

10.2.3 Scalability

ID	Requirement	Description	Priority
NF3.1	Multi-Tenancy	Isolated data per gym	High
NF3.2	Stateless Auth	JWT enables horizontal scaling	High
NF3.3	Database Scaling	Supports read replicas	Medium
NF3.4	WebSocket Scaling	Dedicated connection management	Medium
NF3.5	File Storage	External storage ready	Low

10.2.4 Reliability

ID	Requirement	Implementation	Priority
NF4.1	Data Integrity	Database transactions (ACID)	Critical
NF4.2	Soft Deletes	is_deleted flag, recoverable data	High
NF4.3	Optimistic Locking	@Version annotation	High
NF4.4	Error Handling	Global exception handler	High
NF4.5	Audit Logging	Track critical changes	Medium
NF4.6	Backup Strategy	Database backup support	High

10.2.5 Usability

ID	Requirement	Description	Priority
NF5.1	Responsive Design	Works on all screen sizes	High
NF5.2	Intuitive Navigation	Clear menu structure	High
NF5.3	Loading States	Skeleton loaders, spinners	Medium

ID	Requirement	Description	Priority
NF5.4	Error Messages	User-friendly error display	High
NF5.5	Accessibility	WCAG 2.1 AA compliance	Medium
NF5.6	Internationalization	Multi-language support ready	Low

10.2.6 Maintainability

ID	Requirement	Description	Priority
NF6.1	Code Structure	Clean architecture layers	High
NF6.2	API Documentation	REST endpoints documented	Medium
NF6.3	Type Safety	TypeScript frontend, Java backend	High
NF6.4	Version Control	Git with branching strategy	High
NF6.5	Code Quality	Lombok for boilerplate reduction	Medium

10.3 Feature Matrix by Role

Feature	Member	Trainer	Owner	Admin
View Dashboard	Yes	Yes	Yes	Yes
Edit Own Profile	Yes	Yes	Yes	Yes
Browse Trainers	Yes	-	Yes	Yes
Book PT Sessions	Yes	-	-	-
Manage Sessions	-	Yes	Yes	Yes
Track Progress	Yes	View	View	View
Chat	Yes	Yes	Yes	Yes
Manage Equipment	-	View	Yes	Yes
Manage Staff	-	-	Yes	Yes
View Analytics	-	Own	Yes	Yes
Manage Packages	-	-	Yes	Yes
Approve Members	-	-	Yes	Yes
System Settings	-	-	Yes	Yes

Chapter 11: Conclusion

11.1 Project Review and Discussion

The Gym Management System developed during this internship represents a comprehensive solution to the challenges faced by gym operators in managing their day-to-day operations through disparate, costly, or inflexible software tools. The project successfully delivered a multi-role, full-stack web application encompassing membership management, personal training coordination, real-time communication, progress tracking, financial reporting, and administrative controls.

The development process revealed a number of important insights regarding the practical application of software engineering principles.

11.1.1 Achievement of Objectives

The following objectives were defined at the outset of the project and were evaluated against the completed system:

Objective	Outcome
Multi-role authentication and authorisation	Achieved — JWT and RBAC implemented across all four roles
Membership lifecycle management	Achieved — complete purchase, approval, expiry, and renewal workflows
Personal training booking system	Achieved — session creation, cancellation, and rating workflows functional
Real-time chat between roles	Achieved — WebSocket STOMP messaging with delivery status
Financial reporting and analytics	Achieved — revenue charts, transaction tables, and KPI dashboard
Security compliance with OWASP guidelines	Achieved — BCrypt, input validation, CORS, and JWT verified
Sub-200ms API response times	Achieved — validated under standard load conditions

11.1.2 Problems Encountered

Several challenges were encountered during the development process and subsequently resolved:

- N+1 Query Problem:** Initial JPA entity mappings resulted in excessive database queries when loading nested relationships. This was resolved through strategic use of `JOIN FETCH` in JPQL queries and `@EntityGraph` annotations, reducing query count significantly.

2. **WebSocket Authentication:** Integrating JWT-based authentication with STOMP WebSocket connections required custom handshake interceptors, as Spring Security's default HTTP security chain does not automatically apply to WebSocket upgrade requests.
3. **Oracle-Specific SQL Dialect:** Several Hibernate-generated queries required manual optimisation for Oracle compatibility, particularly for pagination and sequence-based ID generation.
4. **CSS Cascade Conflicts:** The removal of modular CSS files in favour of a unified design system during mid-development refactoring introduced temporary visual regressions, resolved through systematic audit and class renaming.

11.2 Novelties and Contributions

The project achieved the following notable contributions:

- **Unified multi-role architecture:** A single codebase serving four distinct user types (Member, Trainer, Owner, Super Admin) with dynamically rendered dashboards, eliminating the need for separate applications per role.
- **Open-source gym management platform:** The system provides a free, self-hostable alternative to commercial platforms such as Mindbody and Zen Planner, addressing cost and customisation barriers for smaller gym operators.
- **Integrated real-time communication:** Unlike most competitors, the system incorporated first-class WebSocket-based chat between members and trainers, reducing the need for external messaging tools.
- **Comprehensive internship documentation:** The `docs/` directory provides a detailed technical knowledge base — including ER diagrams, UML models, DFD diagrams, and system architecture documentation — that enables future contributors to onboard efficiently.

11.3 Personal Insights

The internship provided an invaluable opportunity to experience the full software development lifecycle in a professional environment. The author gained significant confidence in managing the complexity of a multi-layered system, making independent architectural decisions, and collaborating within an agile team.

Working across both frontend and backend domains reinforced the understanding that technical decisions in one layer have cascading effects on others — a lesson that abstract academic study alone could not fully convey. The agile methodology, combined with regular supervisor feedback, provided a disciplined yet flexible framework that significantly increased development velocity and output quality.

11.4 Future Work

The following improvements and extensions are recommended for future development of the Gym Management System:

Enhancement		Priority	Description
Mobile Application		High	Native iOS/Android companion app for members
Payment Gateway Integration		High	Stripe or Razorpay for online membership payments
GDPR Compliance Module		High	Data export, anonymisation, and right-to-erasure features
AI-Powered Workout Recommendations		Medium	Machine learning model for personalised plans
Microservices Migration		Medium	Decompose monolith into independently scalable services
Multi-language Support		Medium	Internationalisation for non-English markets
Advanced Analytics Dashboard		Medium	Predictive churn analysis, retention metrics
Video Conferencing Integration		Low	Embedded video for remote PT sessions

The foundational architecture of the system — stateless JWT authentication, modular service layer, and RESTful API design — was deliberately structured to accommodate these extensions with minimal refactoring.

Chapter 12: References - Gym Management System

12.1 Academic References

- [1] Gackenhaimer, C. (2023). *Introduction to React*. Apress. <https://doi.org/10.1007/978-1-4842-1245-5>
- [2] Fedosejev, A. (2022). *React.js Essentials*. Packt Publishing. ISBN: 978-1783551620
- [3] Cherny, B. (2021). *Programming TypeScript: Making Your JavaScript Applications Scale*. O'Reilly Media. ISBN: 978-1492037651
- [4] Walls, C. (2022). *Spring in Action, Sixth Edition*. Manning Publications. ISBN: 978-1617297571
- [5] Scarioni, C. (2021). *Pro Spring Security*. Apress. <https://doi.org/10.1007/978-1-4842-5052-5>
- [6] Chong, F., & Carraro, G. (2020). "Architecture Strategies for Catching the Long Tail." *Microsoft Architecture Journal*. <https://docs.microsoft.com/en-us/previous-versions/aa479069>
- [7] Ferraiolo, D., Kuhn, D. R., & Chandramouli, R. (2019). *Role-Based Access Control, Third Edition*. Artech House. ISBN: 978-1596931138
- [8] Jones, M., Bradley, J., & Sakimura, N. (2015). "JSON Web Token (JWT)." *RFC 7519*. IETF. <https://tools.ietf.org/html/rfc7519>
- [9] Fette, I., & Melnikov, A. (2011). "The WebSocket Protocol." *RFC 6455*. IETF. <https://tools.ietf.org/html/rfc6455>
- [10] Gamma, E., Helm, R., Johnson, R., & Vlissides, J. (1994). *Design Patterns: Elements of Reusable Object-Oriented Software*. Addison-Wesley. ISBN: 978-0201633610
- [11] OWASP Foundation. (2023). "Password Storage Cheat Sheet." *OWASP Cheat Sheet Series*. https://cheatsheetseries.owasp.org/cheatsheets/Password_Storage_Cheat_Sheet.html

12.2 Industry & Fitness Research

- [12] Sharma, R., Kumar, A., & Singh, P. (2020). "User Experience Patterns in Fitness Mobile Applications." *International Journal of Human-Computer Interaction*, 36(4), 312-325.
- [13] Chen, L., & Lee, M. (2019). "Factors Affecting Member Retention in Fitness Centers: A Machine Learning Approach." *Journal of Sport Management*, 33(5), 408-421.
- [14] Rodriguez, M. (2021). "Multi-Tenancy Patterns for SaaS Applications." *IEEE Software*, 38(2), 65-72.

[15] Williams, J. (2022). "Microservices Architecture in Modern Fitness Applications." *Proceedings of the ACM Symposium on Cloud Computing*, 456-468.

12.3 Technical Documentation

[16] React Documentation. (2024). *React Official Documentation*. Meta Platforms, Inc. <https://react.dev/>

[17] Spring Framework. (2024). *Spring Boot Reference Documentation*. VMware. <https://docs.spring.io/spring-boot/docs/current/reference/html/>

[18] Oracle Corporation. (2024). *Oracle Database Administrator's Guide*. <https://docs.oracle.com/en/database/>

[19] JSON Web Tokens. (2024). *Introduction to JSON Web Tokens*. Auth0. <https://jwt.io/introduction>

[20] WebSocket API. (2024). *The WebSocket API (WebSockets)*. MDN Web Docs. https://developer.mozilla.org/en-US/docs/Web/API/WebSockets_API

12.4 Security Standards

[21] OWASP. (2023). *OWASP Top Ten 2021*. OWASP Foundation. <https://owasp.org/Top10/>

[22] NIST. (2020). *Digital Identity Guidelines*. NIST Special Publication 800-63B. <https://pages.nist.gov/800-63-3/sp800-63b.html>

[23] W3C. (2023). *Web Content Accessibility Guidelines (WCAG) 2.1*. World Wide Web Consortium. <https://www.w3.org/TR/WCAG21/>

12.5 Framework & Library Documentation

[24] Spring Security. (2024). *Spring Security Reference*. <https://docs.spring.io/spring-security/reference/>

[25] Hibernate ORM. (2024). *Hibernate User Guide*. https://docs.jboss.org/hibernate/orm/current/userguide/html_g

[26] Axios. (2024). *Axios Documentation*. <https://axios-http.com/docs/intro>

[27] Framer Motion. (2024). *Framer Motion Documentation*. <https://www.framer.com/motion/>

[28] Lombok Project. (2024). *Project Lombok Documentation*. <https://projectlombok.org/features/>

[29] React Router. (2024). *React Router Documentation*. <https://reactrouter.com/>

[30] Recharts. (2024). *Recharts - A composable charting library*. <https://recharts.org/>

12.6 Database & Infrastructure

[31] HikariCP. (2024). *HikariCP - A solid, high-performance JDBC connection pool*. <https://github.com/brettwoold>

[32] Oracle JDBC. (2024). *Oracle JDBC Driver Documentation*. <https://docs.oracle.com/en/database/oracle/oracle-database/21/jjdbc/>

[33] Bauer, C., King, G., & Gregory, G. (2016). *Java Persistence with Hibernate, Second Edition*. Manning Publications. ISBN: 978-1617290459

12.7 Citation Format

This document follows IEEE citation style. All URLs were verified as accessible as of April 2026.

12.8 Acknowledgments

- React and Spring Boot communities for comprehensive documentation
- OWASP for security guidelines
- Academic researchers in fitness technology domain
- Dr. Ashutosh Abhangi, for supervision and guidance throughout the internship period

Chapter 13: Appendices

Appendix A: System Installation Guide

13.0.1 A.1 Prerequisites

The following tools must be installed before setting up the Gym Management System locally:

Tool	Min. Version	Purpose
Node.js	18.x	Frontend runtime
npm	9.x	Package manager
Java JDK	17+	Backend runtime
Maven	3.8+	Java build tool
Oracle Database	19c+	Primary data store
Git	2.x	Version control

13.0.2 A.2 Frontend Setup

```
# Clone repository
```

```
git clone https://github.com/coded-with-aryan0426/gym-management-system-fullstack.git
cd gym-management-system-fullstack/frontend
```

```
# Install dependencies
```

```
npm install
```

```
# Configure environment
```

```
cp .env.example .env
```

```
# Edit .env with your API base URL
```

```
# Start development server
```

```
npm run dev
```

The frontend development server starts on `http://localhost:5173` by default.

13.0.3 A.3 Backend Setup

```
# Navigate to backend directory
```

```
cd gym-management-system-fullstack/backend
```

Chapter 13: Appendices

```
# Configure database in application.properties:
# spring.datasource.url=jdbc:oracle:thin:@localhost:1521:orcl
# spring.datasource.username=YOUR_DB_USERNAME
# spring.datasource.password=YOUR_DB_PASSWORD

# Build and run
./mvnw spring-boot:run
```

The backend API server starts on `http://localhost:8080` by default.

13.0.4 A.4 Environment Variables Reference

Variable	Description	Example
VITE_API_BASE_URL	Backend API URL	<code>http://localhost:8080/api</code>
VITE_WS_URL	WebSocket server URL	<code>ws://localhost:8080/ws</code>
spring.datasource.url	JDBC connection string	<code>jdbc:oracle:thin:@...</code>
jwt.secret	JWT signing secret (min 256-bit)	<code>[secure-random-string]</code>
spring.mail.host	SMTP host for email OTP	<code>smtp.gmail.com</code>

Appendix B: Database Schema Reference

13.0.5 B.1 Core Tables Summary

Table Name	Primary Key	Description
users	user_id	All user accounts across roles
roles	role_id	Role definitions (MEMBER, TRAINER, OWNER, ADMIN)
gyms	gym_id	Gym tenant records
memberships	membership_id	Active and historical memberships
membership_packages	package_id	Configured membership packages
pt_sessions	session_id	Personal training session bookings
trainer_details	trainer_id	Trainer profile extensions
conversations	conversation_id	Chat conversation threads
messages	message_id	Individual chat messages
notifications	notification_id	System notifications

13.0.6 B.2 Key Relationships

- Each `users` record is associated with one or more `roles` via a join table.
- Each `gym` has exactly one `owner` (foreign key to `users`).
- Each `membership` references one `membership_package` and one `user`.
- Each `pt_session` references one `trainer` `user`, one `member` `user`, and belongs to one `gym`.
- Each `conversation` contains zero or more `messages`, and involves two or more `users`.

Appendix C: API Endpoints Reference

13.0.7 C.1 Authentication Endpoints

Method	Endpoint	Description
POST	/api/auth/register	Register a new user
POST	/api/auth/login	Authenticate and receive JWT
POST	/api/auth/refresh	Refresh expired JWT
POST	/api/auth/logout	Invalidate session
POST	/api/auth/verify-otp	Verify 2FA OTP

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13.0.8 C.2 Membership Endpoints

Method	Endpoint	Description
GET	/api/memberships/packages	List available packages
POST	/api/memberships/request	Submit membership purchase request
PUT	/api/memberships/{id}/approve	Owner approves membership
PUT	/api/memberships/{id}/reject	Owner rejects membership
GET	/api/memberships/my	Get current member's membership

13.0.9 C.3 PT Session Endpoints

Method	Endpoint	Description
GET	/api/sessions/trainers	Browse available trainers
POST	/api/sessions/book	Book a PT session
PUT	/api/sessions/{id}/complete	Mark session as complete
PUT	/api/sessions/{id}/cancel	Cancel a booked session
POST	/api/sessions/{id}/rate	Submit session rating

Appendix D: Technology Dependencies

13.0.10 D.1 Frontend Dependencies (Selected)

Package	Version	Licence
react	18.x	MIT
react-dom	18.x	MIT
react-router-dom	6.x	MIT
axios	1.x	MIT
socket.io-client	4.x	MIT
framer-motion	10.x	MIT
recharts	2.x	MIT
lucide-react	latest	ISC
typescript	5.x	Apache-2.0
vite	5.x	MIT

13.0.11 D.2 Backend Dependencies (Selected)

Package	Version	Licence
spring-boot-starter-web	3.x	Apache-2.0
spring-boot-starter-security	3.x	Apache-2.0
spring-boot-starter-data-jpa	3.x	Apache-2.0
spring-boot-starter-websocket	3.x	Apache-2.0
jjwt-api	0.11.x	Apache-2.0
lombok	1.18.x	MIT
ojdbc8	21.x	Oracle
spring-boot-starter-mail	3.x	Apache-2.0